

Marriage and Misallocation: Evidence from 70 Years of U.S. History*

Jay Euijung Lee[†]

Stockholm University and Centre for Economic Performance (LSE)

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Abstract

Traditional expectations that married women should be homemakers continue to constrain their labor market participation in many parts of the world. I quantify the aggregate economic consequences of these marriage-specific forces discouraging women’s market work in the historical U.S. context, accounting for selection into marriage and labor force participation. In 1940, married women faced a wedge equivalent to a 44% tax on market wages relative to similar single women; by 2010, it had fallen to 14%. The estimated wedges are strongly related to survey measures of gender-role attitudes, suggesting that they partly reflect gender norms. Had the 1940 wedge persisted, market output in the modern U.S. economy would be 8.7% lower and combined market and home output 5.3% lower. These losses arise from the misallocation of married women’s talent and are amplified by endogenous human capital investment and marriage responses, which reshape the productivity distributions of both women and men.

Keywords: Marriage, gender norms, misallocation, female labor force participation

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[†]email: Jay-Euijung.Lee@su.se

1 Introduction

Despite converging economic lives of men and women—in time use, education, and earnings—in many countries, substantial gender specialization remains between married couples.¹ Under certain forms of the household production function, specialization is required for efficiency (Becker, 1981; Pollak, 2013) and would therefore benefit aggregate productivity. However, recent evidence suggests that specialization within couples does not simply reflect gender-neutral sorting within the household (Hancock et al., 2025).² When specialization reflects gendered frictions rather than efficient allocation, married women with high market potential may remain in home production, resulting in misallocation.

How much do barriers to married women’s labor market participation constrain aggregate output? I first calibrate the wedges required to rationalize married women’s lower labor force participation relative to otherwise similar single women and then quantify their effects on U.S. output between 1940 and 2010. By focusing on barriers specific to *married* women, this paper differs fundamentally from existing work on the aggregate productivity effects of gendered institutions that distinguish between men and women independently of their choices (Hsieh et al., 2019; Erosa et al., 2022; Lee, 2024; Chiplunkar and Goldberg, 2024). Because marriage is an endogenous choice, accounting for selection into marriage is central to the analysis.

The general equilibrium effects of the marriage wedge are far from straightforward. In partial equilibrium—holding individuals’ education and marital status fixed—the implication is simple: a stronger wedge leads married women to work less. In general equilibrium, however, the wedge also affects both marriage and education decisions, making its overall impact more complex. For instance, its effect on women’s incentives to pursue education depends on how it changes the expected probability of working at different education levels. A stronger wedge may discourage labor force participation among married women, but it may also reduce the likelihood of marriage, especially among highly educated women. Because these two channels affect women’s expected probability of working in opposite directions, the overall effect on their labor-market returns to education is ambiguous. The effect on educational attainment is further complicated by the presence of marriage-market returns to education, which may reinforce or offset these labor-market incentives.³

Individuals in my model make three sets of choices over the course of their life cycle. First, they choose their level of education as a forward-looking investment, considering both labor- and

¹Lundberg and Pollak (2007) review this pattern for the United States. Gershuny (2018) document a consistent within-country gender convergence in time use, with men doing more unpaid work and women more paid work, in 12 European countries, South Korea, the United States, and Australia from the 1960s to the 2010s; yet, women still perform nearly 60% of unpaid work even where convergence is most advanced.

²Hancock et al. (2025) show that the allocation of home production is not gender-neutral even after accounting for earnings differences, and that matching these patterns requires gendered frictions in the assignment of household tasks.

³Section 3.7 describes these channels in more detail.

marriage-market returns. Second, they enter a frictionless transferable-utility marriage market in the style of [Becker \(1973\)](#), where individual types are defined by education. They decide which spousal type to marry or whether to stay single, and then draw a children-composition category (the number and ages of children) according to match-specific empirical probabilities. Fertility is not modeled as a structural choice for tractability, as I always net out the effect of children when isolating the marriage wedge.⁴ Third, individuals draw market and home abilities, and households make the dichotomous labor supply choice of whether each individual works in the market or in home production. I focus on the binary work decision instead of hours, because the growth in married women’s total market work hours between 1940 and 2010 is almost entirely driven by the extensive margin of participation rather than the intensive margin of hours conditional on working.⁵ The wedge is modeled as a disutility for a married couple when the wife works in the market ([Akerlof and Kranton, 2000](#); [Bertrand et al., 2015](#)), acting like a tax that lowers the value of her market wage.

What does the marriage wedge reflect? I interpret traditional gender norms within marriage as an important component. The wedge is not, however, a pure measure of norms: it is calibrated as a composite force that makes married women’s labor force participation differ from that of similar⁶ single women, beyond what can be explained by wage differentials. Within this composite force, norms can enter in a variety of ways: e.g., women’s preferences to conform to traditional roles as wives, higher valuation of staying home among married women relative to single women, differential non-wage treatment by firms of married and single women, and men’s preferences for wives who are more productive at home. In support of the norms interpretation, the model-derived wedges are strongly correlated with survey measures of attitudes toward gender roles in marriage, both across U.S. states and over time within states. To address competing explanations, I also show that the calibrated wedges are robust across alternative modeling and calibration approaches that account for other important differences between married and single women unrelated to norms, including income effects from husbands’ earnings, the tax treatment of marriage ([Guner et al., 2012](#); [Gayle and Shephard, 2019](#)), and differences in childbearing prospects.

The model is matched to education, marriage, and labor-force participation patterns in the U.S. decennial census, decade by decade, to trace how the magnitude of the marriage wedges changes over time.⁷ I then use the calibrated model to study how aggregate output would change if the

⁴I net out the effect of children when comparing married and single women in the motivating evidence (Figures 1 and 2) and in the calculation of the marriage wedge. Nonetheless, the match-specific empirical probabilities of children composition are calibrated separately for each decade, allowing the model to capture shifts in fertility patterns—such as the baby boom and subsequent decline, and changes in how fertility relates to income, female labor force participation, and female education ([Doepke et al., 2023](#)).

⁵See Appendix D for a visual illustration and a formal decomposition.

⁶I.e., matched on education and child composition—such as college-educated women with one child under age 5.

⁷Throughout this paper, I treat cohabiting couples as married, since they are subject to similar—albeit likely weaker—gender roles as legally married couples. I define married couples as those with “probable” spouses living in the same household, as identified in the IPUMS census data ([Ruggles et al., 2024](#)), which includes cohabiting cou-

marriage wedge were altered. These counterfactuals capture both the direct effects on labor supply and the indirect effects operating through forward-looking marriage and education choices.

I find that married women faced a wedge equivalent to 44% of the market wage in 1940, declining to 14–15% by 2000 and 2010. Had the wedge remained at its 1940 level, aggregate market output in 2000 would have been 8.7% lower, while aggregate total (market and nonmarket) output would have been 5.3% lower. These losses amount to 12.6% of the growth in market output and 8.1% of the growth in total output between 1940 and 2000. To gauge the magnitude of these results, I compare them to the output effect of another counterfactual: if married women of 2000 made labor force participation choices based on 1940 wages and home productivities. Relative to this benchmark, the 1940 marriage wedge generates nearly 70% as large an output loss. This comparison shows that the decline in the marriage wedge is quantitatively important even relative to large historical advances in women’s economic incentives to work in the market.

Two forces drive these output effects: lower aggregate human capital and greater misallocation of married women’s market talent. Importantly, a substantial share of the output loss arises from declining market productivity among men—a general equilibrium effect that would be missed by a partial equilibrium analysis holding education and marriage choices fixed. The share attaining the highest education level—postgraduate studies—falls by 10.7% among men and 14.1% among women, while average years of schooling fall by 2.0% and 2.6%, respectively. This partly reflects the decline in marriage-market returns to education as marriage becomes less attractive: the marriage rate falls by one-quarter, with especially large declines among highly educated women. For women, lower labor-market returns to education reinforce the decline in schooling caused by weaker marriage incentives. Married women’s labor-force participation and cumulative market earnings fall by 34.1% and 28.2%, respectively, reflecting both greater misallocation under a higher marriage wedge and more negative selection into marriage.

The counterfactual also implies that the decline in the marriage wedge between 1940 and 2010 accounts for several well-documented trends in the U.S.: a) rising labor force participation among married women, b) increasing wives’ share of household income, c) rising education among both men and women, especially women, and d) increasingly positive selection into marriage by education (Greenwood et al., 2016; Bar et al., 2018).⁸ In particular, it accounts for 38% of the rise in the labor force participation of married women. To the extent that the marriage wedge captures traditional gender norms, these results suggest that the weakening of those norms played a quantitatively important role in these broader changes.

The U.S. historical experience is informative beyond this setting. Many countries are undergo-

ples from 1990. Appendix H.2 examines the sensitivity of the calibrated marriage wedges to defining marriage solely by legal marital status; the resulting wedges are slightly higher, as expected.

⁸The decline in the marriage wedge cannot explain the long-term decline in U.S. marriage rates; a separate parameter captures the diminishing attractiveness of marriage.

ing similar transitions: married women’s labor force participation is catching up with that of single women, while attitudes toward gender roles are becoming less conservative.⁹ These patterns appear not only among richer OECD economies but also in low- and middle-income countries in Eastern Europe and Latin America. At the same time, one in ten countries still has a lower female labor force participation rate than the United States did in 1940 (ILO, 2024),¹⁰ indicating that the conditions studied in this paper remain current in parts of the world today. Moreover, gender attitudes can evolve unevenly or stall, and minority support for traditional views can shift perceived social norms when people overestimate their prevalence.¹¹ The historical U.S. analysis therefore provides a benchmark for the potential output gains in economies moving away from traditional gender roles, as well as the output stakes if such progress were to reverse.

A large literature pioneered by Restuccia and Rogerson (2008) and Hsieh and Klenow (2009) studies the aggregate productivity implications of various forms of misallocation. A growing number of papers examine gender-based distortions in the allocation of talent—discrimination in educational and labor markets distorting occupational choice (Hsieh et al., 2019); differences in non-market time using married couples only, without considering marriage as a choice (Erosa et al., 2022); entry barriers to the non-agricultural sector (Lee, 2024); and barriers to entrepreneurship (Chiplunkar and Goldberg, 2024). I add to this literature by focusing on a distortion that is conditional on marriage rather than one that applies uniformly across women. Endogenizing marriage makes selection into exposure to the distortion part of the equilibrium and opens an additional channel through marriage-market returns to education.

I also contribute to the literature at the intersection of macroeconomics and family economics, which emphasizes that marriage and family formation are critical for understanding various aggregate outcomes—business cycles, income inequality, fertility, labor supply, savings, long-run development, and political institutions (see Doepke and Tertilt (2016) for an overview). Yet less is known about how aggregate output is affected by marriage-specific forces that shift women between market work and home production. This paper quantifies the output consequences of this margin and finds sizable effects.

My theoretical contribution is to build a tractable general equilibrium model with closed-form solutions, while maintaining flexibility to account for key secular trends in the gender wage gap,

⁹See Appendix C for figures documenting these trends worldwide. The appendix also shows a positive cross-country correlation between conservative gender attitudes and the married–single gap in women’s labor-force participation, underscoring the broader relevance of marriage-specific gender roles beyond the United States.

¹⁰In light of the U-shaped pattern of female labor force participation over the cycle of economic development (Goldin, 1995; Mammen and Paxson, 2000; Ngai et al., 2024), found when unpaid family labor is included, I should clarify that this ILO statistic pertains only to paid employment and self-employment.

¹¹Bursztyn et al. (2023) document that people tend to misperceive prevailing social norms about whether “women should have the freedom to work outside the home” in the United States and many other countries, placing disproportionate weight on minority views.

gender differences in home productivity, the propensity of marriage, assortativeness of marriage matching by education, fertility patterns, and educational attainment. The novelty lies in extending a tractable selection framework—commonly used in the trade literature (Eaton and Kortum, 2004; Hsieh et al., 2019) to study *individual* decisions—to *household* decision-making, and integrating it with standard models of educational choice and marriage market matching (Choo and Siow, 2006; Chiappori et al., 2017; Chiappori et al., 2018). The resulting closed-form solutions clarify how education, marriage, and labor supply decisions interact and make the parameter identification procedure fully transparent. While recent structural work on marriage markets and family labor supply (e.g., Goussé et al., 2017; Gayle and Shephard, 2019; Calvo et al., 2024) incorporates richer mechanisms, estimating a large number of parameters and comparing them consistently across many time periods can be challenging because of their complexity. In contrast, the combination of tractability and flexibility in my framework is particularly well-suited for tracing the evolution of marriage wedges decade by decade over 70 years.

Lastly, I contribute to a large body of both macro and micro work that seeks to explain the dramatic rise in married women’s labor force participation in the U.S. Existing explanations can be broadly categorized into two branches: technological progress and cultural change. The technology branch includes the invention of birth control pills (Goldin and Katz, 2002), technological advances in housework (Greenwood et al., 2005), gender-biased technological progress (Heathcote et al., 2010), medical progress in pregnancy-related conditions (Albanesi and Olivetti, 2016), and the rise in services driven by structural transformation and marketization (Ngai and Petrongolo, 2017). The culture branch includes greater acceptance of working wives by men (Fernández et al., 2004), changes in social attitudes toward married women working (Fernández, 2013), reforms to divorce laws (Fernández and Wong, 2014), and falling gender wage discrimination (Jones et al., 2015). I contribute to the latter branch in two ways. First, I show that the marriage wedges I recover are closely related to cultural norms governing women’s roles within marriage. Second, I quantify the aggregate importance of these wedges in accounting for the rise in married women’s labor force participation over the past several decades.

The paper proceeds as follows. Section 2 presents empirical evidence motivating a wedge that distinguishes the labor supply of married and single women. Section 3 develops a model that formalizes this wedge and explores its implications for education and marriage decisions. Section 4 describes the data and calibration procedure and discusses the resulting parameter estimates. Section 5 uses the model to quantify the aggregate effects of changes in the wedge. Section 6 concludes.

2 Motivating Facts

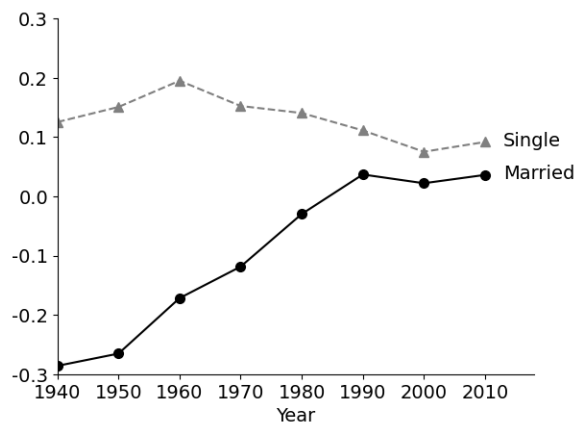
This section presents three descriptive facts that motivate the focus on marriage-specific forces in women’s labor supply. First, married women’s labor force participation, even after accounting for demographics and children, remains lower than that of otherwise similar single women. The gap, however, has narrowed over time. Second, women exhibit large, sudden drops in labor force participation after marriage, while men do not; these drops are smaller among those marrying later. Third, survey attitudes toward gender roles in marriage became substantially less traditional over the same period. Together, these facts motivate a marriage wedge on women’s market work that weakens over time, where gender norms may be an important component.

“Unexplained” female labor force participation I compare the path of the “unexplained” labor force participation (LFP) of married and single women over time. By unexplained LFP, I refer to residuals from regressing the LFP indicator on standard, commonly observed individual characteristics X : age, education, race, and number of children dummies.

$$LFP_{it} = X_{it}\beta + \epsilon_{it}$$

After running the regression on all females aged 25-54 between 1940 and 2010 in the U.S. decennial census, I take the weighted average of the residuals by marital status and decade.

Figure 1: Residualized Female Labor Force Participation, by Marital Status



Note: This figure compares the labor force participation rates of single (never married, separated, divorced, widowed) to married women between the ages of 25 and 54 in 1940 and 2010, in the United States. The participation rates are residualized for age, education, race, and number of children dummies.

Figure 1 shows that residualized LFP rose sharply for married women over this period, catching up to the relatively stable level for single women. Two points are important. First, married women’s labor supply differs substantially from that of single women with similar observable characteristics.

Second, this difference has narrowed over time. Importantly, because the comparison controls for the number of children, the married-single gap is not simply a children margin—the dominant factor setting marrieds apart from singles in the literature. Technological and institutional advances related to childbearing and child-rearing therefore cannot explain the catch-up. These patterns motivate an evolving marriage-specific wedge in women’s labor supply.

Shift in work patterns upon marriage To further corroborate the marriage wedge, I analyze how women adjust their time use immediately upon marriage. I contrast these shifts with those observed among men to shed light on gender roles within marriage.

I follow the event-study approach of [Kleven et al. \(2019\)](#). For this exercise, I use the Panel Study of Income Dynamics 1968-2015 ([Panel Study of Income Dynamics, public use dataset, 2024](#)), a panel data of nationally representative individuals of the United States. Men and women often live in different family arrangements prior to marriage—for example, women may be more likely to reside in their parental home, while men may be more likely to live alone. To minimize the influence of other adult household members on time use before marriage, I restrict the sample to individuals who transition from being unmarried household heads to being married with live-in spouses. The survey treats permanent cohabitation as marriage.¹²

I define event times to be years relative to marriage, such that event time 0 refers to the first year in which an individual switches from single to married. I regress

$$Y_{ist} = \sum_{j \neq -1} \alpha_j \cdot \mathbb{1}(t = j) + \sum_k \beta_k \cdot \mathbb{1}(age_{is} = k) + \sum_y \gamma_y \cdot \mathbb{1}(s = y) + X_{ist} \delta + \nu_{ist}$$

This regression tracks how the outcome variable Y of individual i in calendar year s at event time t changes as a function of event time, while controlling for life-cycle dynamics via the age dummies and time trends via the year dummies. I consider two outcome variables: labor force participation and weekly housework hours.¹³ X includes additional control variables—the number of children and the age of the youngest child—to distinguish the evolution of outcomes around marriage from that around childbirth. Regressions are run separately by gender and, to examine whether “marriage effects” change over time, across two distinct marriage cohorts: earlier and later marriages.¹⁴

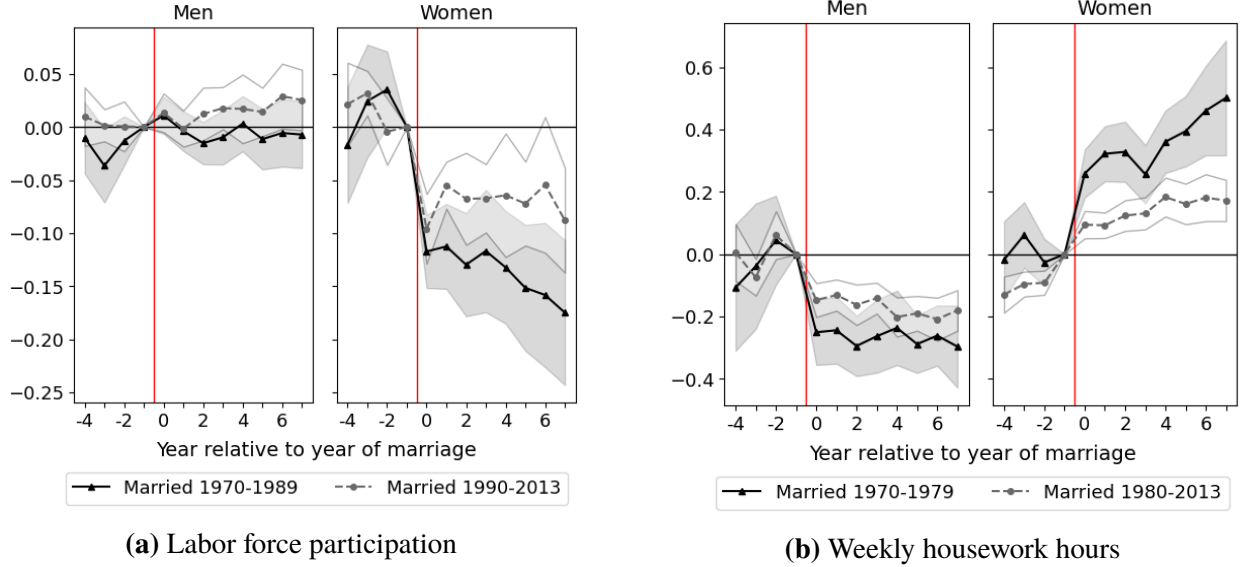
¹²Additional sample restrictions are: (i) only including years from three years before marriage (the year marital status changes from 0 to 1) to eight years after, (ii) individuals with only one marriage recorded for a clean event study design, and (iii) restricting marriages to those occurring between 1970 and 2013.

¹³For housework hours, the survey asks, “About how much time does (he/she) (do you) spend on this housework in an average week—I mean time spent cooking, cleaning, and other work around the house?”

¹⁴The earlier marriage cohort is chosen to capture marriages near the beginning of the sample period, subject to data availability. For housework hours—measured from 1969—I define marriage cohorts as individuals married 1970–1979 (earlier) and 1980–2013 (later). In contrast, employment status of wives of household heads is only measured from 1979. Thus, for labor force participation, I define the marriage cohorts as those married 1970–1989 (earlier) and 1990–2013 (later) to ensure sufficient pre-marriage data for the earlier cohort.

The event-time coefficients ($\hat{\alpha}_t$) are then normalized by $\mathbb{E}[\tilde{Y}_{ist}|t]$, where $\tilde{Y}_{ist} \equiv \sum_k \hat{\beta}_k \cdot \mathbb{1}(age_{is} = k) + \sum_y \hat{\gamma}_y \cdot \mathbb{1}(s = y) + X_{ist} \hat{\delta}$. This normalization expresses $\hat{\alpha}_t$ relative to the level of the outcome that would be predicted at event time t absent any event-time effects. Figure 2 plots $\hat{\alpha}_t / \mathbb{E}[\tilde{Y}_{ist}|t]$.

Figure 2: Event Study—Labor Force Participation and Housework Hours Around Marriage



Note: This figure plots $\hat{\alpha}_t / \mathbb{E}[\tilde{Y}_{ist}|t]$, the evolution of Y around the year of marriage relative to its predicted level in the absence of marriage. The red vertical line plots the timing of marriage. Individuals are unmarried household heads without any live-in partners in the years to the left of the red line, and they are married with live-in spouses in the years to the right of the red line. 95% confidence intervals are reported, based on bootstrapped standard errors computed using 100 resamples at the individual level to account for the panel structure of the data.

Panel (a) plots the results for labor force participation. It illustrates a sharp drop in women's labor force participation immediately upon marriage, and more so for the earlier marriage cohort (-11.7% drop relative to the predicted level in the absence of marriage) than the later one (-9.6%). In contrast, there are no sharp changes for men in either marriage cohort. Panel (b) plots the results for weekly housework hours. There is a sharp 22.6% increase in women's weekly housework hours immediately upon marriage for the earlier marriage cohort, whereas the increase for the later cohort is more muted. For men, there is a sharp decrease upon marriage that is greater for the earlier cohort.

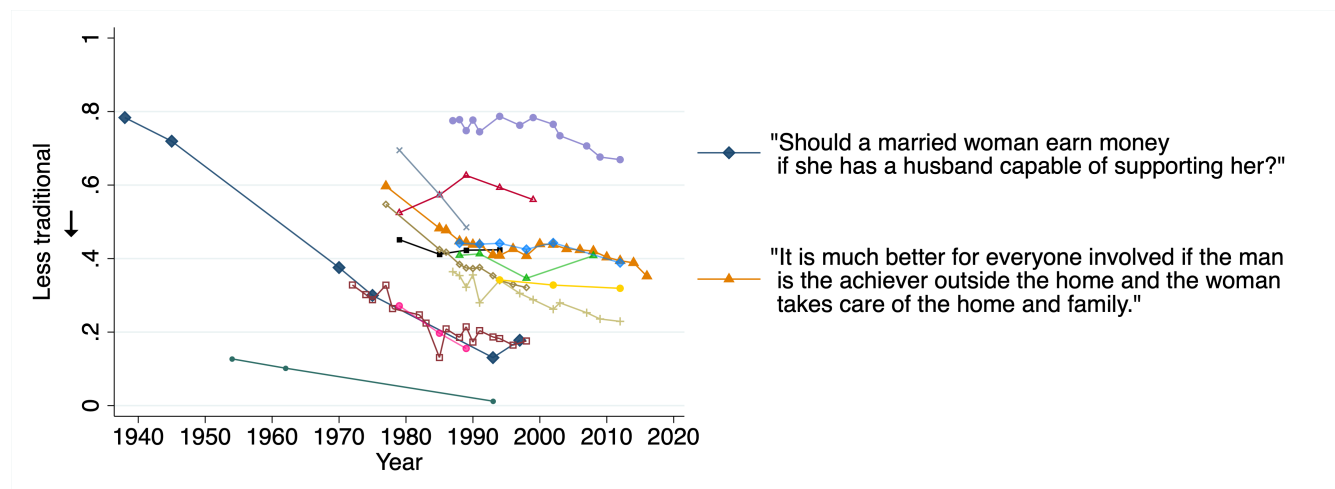
The stark divergence in labor force participation and housework hours between men and women upon marriage shows that marriage plays a pivotal role in shaping gendered divisions of labor, particularly in earlier marriage cohorts. Since the regression controls for the number of children and the age of the youngest child,¹⁵ the results are consistent with gendered expectations attached to

¹⁵The regression results remain similar if instead of controlling for these, I restrict the sample to couples who had no childbirths in the first three years of marriage.

marriage itself—separate from parenthood. This interpretation aligns with Figure 1, where married women’s labor force participation remains lower than that of single women even after adjusting for children. The attenuated shifts observed for women in the later marriage cohort may partly reflect the increasing marketization of household chores over time, such as the greater availability of household appliances (Greenwood et al., 2005). However, combined with the smaller decline in housework hours for men in the later cohort, they also suggest a shift in gender roles in marriage.

Attitudinal survey trends The final motivating fact directly illustrates the weakening of gender roles within marriage over time, a pattern documented in prior work (e.g. Fernández (2013) for the U.S., 1938-1998; Goussé et al. (2017) for the UK, 1991-2008). To capture a more comprehensive view of attitudinal trends that covers every decade from 1940 to 2010, I compile attitudinal survey data from the ROPER polls database. From the set of survey questions returned by keyword searches for “women” and “gender,” I select those related to gender roles within marriage that were asked repeatedly over multiple time periods. For comparability across questions, responses are rescaled to lie between 0 and 1, with 1 indicating the most traditional answer and 0 the least.

Figure 3: Trend in Attitudinal Survey Answers



Note: This figure plots the yearly weighted mean of responses to attitudinal survey questions related to gender roles within marriage from the ROPER polls database. The precise survey questions are shown for two questions that have been asked for the longest periods and together span the entire time frame. A complete legend with all 14 survey questions is provided in Appendix Figure A1. All surveys are based on nationally representative samples of adults or, in some cases, of men and women separately.

Figure 3 shows a clear trend toward less traditional attitudes over time, though the pace of change may be slowing in more recent decades. The variation in these answers across states and decades is used in Section 4.3 to assess the norms interpretation of the marriage wedges.

3 Model

The motivating facts presented in Section 2 show that labor supply differs systematically between married and single women, and that this gap has narrowed over time. They also suggest that gender roles within marriage may be an important source of this difference. In this section, I develop a model that formalizes a marriage wedge on women’s market work and studies how it shapes labor supply, education, and marriage decisions. Because the wedge applies specifically to married women, modeling selection into marriage is essential. The model therefore incorporates marriage matching based on educational attainment, the primary observable trait along which couples sort and the key predictor of gender attitudes.¹⁶

Timing The decisions of individuals are divided into three stages. In the first stage, individuals choose their level of education as a forward-looking investment, balancing the returns to education in the labor and marriage markets with the cost of education, as in [Chiappori et al. \(2009\)](#).

In the second stage, they enter a frictionless transferable utility (TU) marriage market ([Becker, 1973](#)), where types of individuals equal their education levels chosen previously, and decide on which spousal type to get married to or to stay single. The resulting match is thus characterized by the education levels of the husband and wife for married couples and by one’s own education level for singles. Households then exogenously get assigned children composition categories based on the number of children under the age of 5 and under the age of 18 in the household, according to match-specific empirical probabilities. Fertility is not modeled as a structural choice for tractability, as I always net out the effect of children when measuring the marriage wedge.

Individuals subsequently enter the third and last stage, each characterized by group: the tuple of (gender, marriage match, children composition). After individuals draw idiosyncratic market and home abilities, households make the dichotomous labor supply choice of whether to work in the market or in home production for each individual, taking as given a) the group-specific market wage that the representative firm of the economy pays, b) the group-specific value of home production, and c) the group-specific marriage wedge that lowers the value of the wife’s market work. I focus on the binary work decision because the growth in married women’s total market work hours between 1940 and 2010 is almost entirely driven by the extensive margin of participation rather than the in-

¹⁶The model does not include attitudes toward gender roles as part of individuals’ types in the marriage market. Consequently, it does not capture, for instance, whether more traditional men or women are more likely to marry, or whether individuals match based on shared family values. However, this type of matching appears empirically limited. I show in Appendix Table A1 that attitudes toward gender roles are no different by marital status and are instead predicted by far the best by education, using U.S. attitudinal survey data described in Section 2. Similarly, [Goussé et al. \(2017\)](#), who explicitly incorporate family values into individual types, find no evidence of homophily in family values in Britain between 1991 and 2008. In fact, they find no evidence of complementarities in any pairwise interaction of market wages (labor productivity), family values, and education of men and women—except for homophily in education.

tensive margin of hours conditional on working (see Appendix D). After the labor supply decisions are made, households consume and realize utilities.

Since I solve the model backwards, I describe each stage in greater detail, in reverse order.

3.1 Economic utilities and optimal labor supply choices

Individual utilities consist of an economic and a predetermined noneconomic component. The *economic* component is characterized by the utility functions below, adapted from Chiappori et al. (2018).¹⁷ The economic gains from marriage stem from two sources: economies of scale in consuming public goods and risk sharing between spouses against uncertain future public and private consumption.

Married Individuals Consider a married household composed of husband m and wife f . Individual $i \in \{m, f\}$ gets the following utility:

$$u_i(Q, C_i, L_f) = \ln(Q) + \ln(C_i - \tau_i w_f L_f) \quad (1)$$

where C_i is the consumption of the private good, and Q is the consumption of the public good (e.g. expenditures on housing, children, heating). $L_i \in \{0, 1\}$ denotes the labor market participation, and w_i the market wage, of individual i . Critically, married individual i gets disutility when the wife works in the market. The parameter τ_i measures the value of the disutility as a proportion of the market income brought home by the wife.¹⁸ The focus on a wedge affecting the wife's LFP, but not the husband's, is motivated by the absence of any sharp change in men's LFP upon marriage—contrary to sharp drops for women—shown in the previous section (Figure 2).

From an ordinal perspective, this utility belongs to Bergstrom and Cornes (1983)'s Generalized Quasi Linear (GQL) family. Hence, for any realization of family income, it satisfies the transferable utility (TU) property. Under TU, utility can be transferred between spouses at a fixed rate of exchange, and so for Pareto-efficiency, a couple acts as a single decision unit that maximizes the joint marital output. As the set of Pareto efficient allocations is an ordinal concept, any cardinalization of u can be used for the definition of joint marital output. I use $\exp(u_i)$ as the cardinalization of i 's preferences. Then, conditional on the couple's labor choices, any Pareto efficient allocation maximizes the sum of the spouses' exponential utilities, $\exp(u_m) + \exp(u_f)$.

¹⁷A more general formulation of the utility function that is consistent with my model is described in Appendix E.1.

¹⁸The multiplicative relationship between τ_i and w_f implies that, for a given value of τ_i , the disutility is greater in absolute terms for households with higher-earning wives—i.e., the marriage wedge is most costly in these households. One rationale is that higher-earning women often work more paid hours, and the disutility may increase with time spent in market work.

Therefore, conditional on labor market participation choices, a married couple solves

$$\begin{aligned} \max_{Q, C} & Q(C - \tau w_f L_f) \\ \text{s.t.} \quad & pQ + C = w_m L_m + w_f L_f + h_m(1 - L_m) + h_f(1 - L_f) \end{aligned} \quad (2)$$

$C \equiv C_m + C_f$ denotes total expenditure on private goods, $\tau \equiv \tau_m + \tau_f$ is the couple's joint disutility from the wife working in the market, and p is the price of the public good relative to the private good (the numeraire). Moreover, h_i refers to the home productivity of individual i .¹⁹ The solutions to the maximization problem given by (2) are

$$\begin{aligned} Q &= \frac{w_m L_m + (1 - \tau)w_f L_f + h_m(1 - L_m) + h_f(1 - L_f)}{2p} \\ C &= pQ + \tau w_f L_f \end{aligned} \quad (3)$$

I now describe the intra-household allocation, before market earnings and home productivities are realized. Efficient sharing of risks against the uncertainty of earnings and home productivities implies that the ratio of marginal utilities of private consumption is constant and equal to the Pareto weight (μ),²⁰ determined endogenously in the marriage market:

$$\frac{\partial u_m}{\partial C_m} = \mu \frac{\partial u_f}{\partial C_f}$$

The resulting indirect utilities are:

$$v_m = 2 \ln Q + \ln p + \ln \frac{1}{1 + \mu}, \quad v_f = 2 \ln Q + \ln p + \ln \frac{\mu}{1 + \mu} \quad (4)$$

The total economic utility generated from this marriage then is

$$v = v_m + v_f = \check{v} + \ln \frac{\mu}{(1 + \mu)^2} \quad (5)$$

where $\check{v} \equiv 4 \ln Q + 2 \ln p$. It is straightforward that the farther the wife's Pareto weight μ is from 1, the husband's, the smaller the total economic utility of the couple.

The optimal labor choices, which maximize Q , are

$$\begin{aligned} L_m^* &= \mathbb{1} [w_m \geq h_m] \\ L_f^* &= \mathbb{1} [(1 - \tau)w_f \geq h_f] \end{aligned} \quad (6)$$

In the labor choice of married women, τ enters as a marriage wedge: a married woman's market

¹⁹It might seem odd that according to the budget constraint, private and public goods can be bought with "income" from home production. However, Appendix E.2 shows that the utility maximization problem remains identical if I divide goods into market goods and home-produced goods, and have two separate budget constraints for each.

²⁰Browning et al. (2014) characterize intra-household allocations under efficient risk-sharing.

wage is valued below its face value, as if it were taxed.²¹ One feature that stands out in solution (6) is the independence of the husband's and wife's labor supply choices.²² While this feature makes studying the selection into working in the labor market by either the husband or the wife easy, it is not realistic. I consider an extension of the model that I refer to as the “extension with income effects” in Appendix E.4: assuming men always work, I relax the independence so that the labor supply decisions of women are responsive to their husbands' market earnings. I show later in Section 4.3 that the benchmark model with independent spousal decisions performs well up to 2000.

The flexibility of the model allows for another extension (Appendix E.5) in which married men also face wedges, potentially reflecting a “breadwinner” norm: the household incurs a cost when the husband is a homemaker.²³

Single Individuals To distinguish from the married case, I use the hat symbol for singles. The economic utilities of singles follow the same structure as those of married couples, except that they are not subject to the marriage wedge. Consequently, single women do not incur disutility from participating in the labor market.

A single individual i solves

$$\begin{aligned} \max_{\hat{Q}_i, \hat{C}_i, \hat{L}_i} \quad & \hat{u}_i(\hat{Q}_i, \hat{C}_i) = \ln(\hat{Q}_i) + \ln(\hat{C}_i) \\ \text{s.t.} \quad & p\hat{Q}_i + \hat{C}_i = w_i\hat{L}_i + h_i(1 - \hat{L}_i) \end{aligned} \quad (7)$$

The resulting indirect utility is

$$\hat{v}_i = 2 \ln \hat{Q}_i + \ln p \quad (8)$$

and the optimal labor choice that maximizes $\hat{Q}_i$ is

$$\hat{L}_i^* = \mathbb{1} [w_i \geq h_i] \quad (9)$$

Market Wage and Home Productivity Market wages and home productivity depend on idiosyncratic market and home abilities, as well as components common to groups defined by gender, marriage match, and children composition. I abstract from fertility decisions, and instead assume that children composition is drawn exogenously according to match-specific empirical probabilities. Hence, the probability that a (husband education level q , wife education level r) match has

²¹The marriage wedge is a catch-all term for factors that make married women's labor force participation differ from that of similar single women beyond wage differentials. Section 4.3 discusses the different factors captured by the wedge and the evidence linking it to gender norms.

²²Relatedly, as $\frac{\partial^2 \ln Q}{\partial w_m \partial w_f} = 0$, the model does not build in any assortative matching on market earnings.

²³Appendix H.1 presents the calibration of these wedges for married men. They decrease over time like their female counterparts. However, they come with caveats, which I discuss.

a children composition $\mathcal{K}$, denoted as $d^{qr}(\mathcal{K})$, is simply found from corresponding shares in the data.²⁴

Individual i of gender g in a (q, r) match, with children composition $\mathcal{K}$, earns market wage

$$w_i = \bar{w}_g^{qr}(\mathcal{K})\epsilon_i^w, \quad g \in \{M, F\}$$

where $\bar{w}_g^{qr}(\mathcal{K})$ is the market wage per unit of effective labor for i 's group, and ϵ_i^w is i 's market ability. $\bar{w}_g^{qr}(\mathcal{K})$ is allowed to differ flexibly across groups and are treated as exogenous in the current model—implying, for example, that marriage or childbearing causally affects market productivity. In practice, however, calibrated group wages may also reflect selection on unobserved ability not captured by the model.²⁵ While the source of wage differences—causal or selection—does not affect the model's first goal of measuring the marriage wedge, it does affect its second goal of counterfactual analysis if selection is likely to change in counterfactual scenarios. Accordingly, in the counterfactual exercises in Section 5, I consider adaptations based on standard assumptions or empirical decompositions from the literature that separate causal effects from selection.

The home production value of individual i is given by

$$h_i = \bar{h}_g^{qr}(\mathcal{K})\epsilon_i^h, \quad g \in \{M, F\}; \quad \text{and} \quad \bar{h}_F^{qr}(\mathcal{K}) = \bar{h}_F^r(\mathcal{K}).$$

$\bar{h}_g^{qr}(\mathcal{K})$ represents the group-specific component of home productivity. For women, I further assume that it is unaffected by marital status or spousal education, conditional on own education and children composition. The assumption allows me to disentangle marriage wedge parameters from home productivity parameters. Multiple strands of evidence from time use analyses and home production function estimation suggest that this assumption might be reasonable.²⁶ Nonetheless, I explore the implications of relaxing this assumption in an extension of the model (Appendix E.4), in which non-labor income—particularly husband's earnings for married women—influence the relative productivity of home production versus market work.²⁷

²⁴A sizable share of singles have children in the data, because singles include never-married, divorced, separated, and widowed individuals. This grouping of singles is equivalent to assuming that divorces, separations, and widowhoods occur via shocks exogenous to the schooling years of the couple.

²⁵For example, the extensive literature on returns to schooling pioneered by [Griliches \(1977\)](#) highlights the correlation between schooling and unobserved abilities.

²⁶First, [Ramey and Ramey \(2010\)](#) estimate a minuscule coefficient on household income in explaining mothers' childcare time once their own education and the number and ages of children are controlled for—even among mothers who do not work for pay—suggesting that income effects stemming from a husband's earnings are likely to be small. They use the U.S. Bureau of Labor Statistics time-use surveys, 2003–2008. Second, home production time of married couples in American Time Use Surveys 2003–2006 have been found not to depend on spousal relative wages ([Connelly and Kimmel, 2009](#)) or relative education ([Gupta and Stratton, 2008](#)). Third, [Goussé et al. \(2017\)](#) report no significant differences in home production function parameters by marital status for women, using the British Household Panel Survey, 1991–2008. They also find that even own education has little effect on home production specialization parameters within couples, suggesting spousal education, too, is unlikely to be important.

²⁷This extension requires another assumption: that married men always work.

I assume that market abilities ϵ^w and home production abilities ϵ^h are drawn independently and identically from the Fréchet distribution with shape parameter θ , after the marriage matching stage. The Fréchet distribution has been shown to have a similarly good fit with earnings as more commonly used distributions such as the lognormal (Pindado et al., 2017). The cumulative distribution functions for these abilities are $F(\epsilon^w) = F(\epsilon^h) = F(x) = \exp \{-x^{-\theta}\}$.

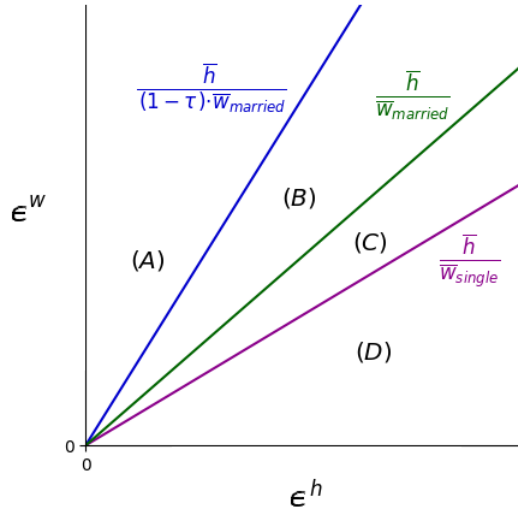
From the property of Fréchet distributions, the probability that a woman in a (q, r) match with children composition $\mathcal{K}$ works in the labor market is given by:

$$P_F^{qr}(\mathcal{K}) \equiv \mathbb{P}((1 - \tau^{qr}(\mathcal{K}))\bar{w}_F^{qr}(\mathcal{K})\epsilon^w > \bar{h}_F^r(\mathcal{K})\epsilon^h) = \frac{[(1 - \tau^{qr}(\mathcal{K}))\bar{w}_F^{qr}(\mathcal{K})]^\theta}{[(1 - \tau^{qr}(\mathcal{K}))\bar{w}_F^{qr}(\mathcal{K})]^\theta + [\bar{h}_F^r(\mathcal{K})]^\theta} \quad (10)$$

The maximum likelihood estimator of this probability is the labor force participation rate of the women in this group. Equation (10) is useful for calibrating parameters in Section 4. Another useful implication of Fréchet abilities is that the average wage of women working in the market is:

$$\begin{aligned} \text{avgwage}_F^{qr}(\mathcal{K}) &= \bar{w}_F^{qr}(\mathcal{K}) \mathbb{E}[\epsilon^w | (1 - \tau^{qr}(\mathcal{K}))\bar{w}_F^{qr}(\mathcal{K})\epsilon^w > \bar{h}_F^r(\mathcal{K})\epsilon^h] \\ &= \bar{w}_F^{qr}(\mathcal{K}) \left(\frac{1}{P_F^{qr}(\mathcal{K})} \right)^{\frac{1}{\theta}} \Gamma\left(1 - \frac{1}{\theta}\right) \end{aligned} \quad (11)$$

Figure 4: Sorting Across Market Work and Home Production of Married and Single Women



Note: This figure shows how labor is allocated between market work and home production for different combinations of market abilities ϵ^w and home abilities ϵ^h , for married and single women with the same education level and children composition (in simplified notation).

- (A): For a married woman to work in the market, her relative talent in market work must be very high.
- (B): If the marriage wedge τ were removed, more married women would engage in market work.
- (C): Along with (A) and (B), (C) covers the ability combinations of single women working in the market.
- (D): Single women who work at home have very high relative talent in home production.

Additionally, Figure 4 illustrates how sorting across market work and home production by market and home abilities occurs for married and single women.

3.2 Marriage market

Marriage matching depends on the expected economic and noneconomic value of each match.

Following Choo and Siow (2006)'s matching under transferable utility (TU) with random preferences, consider an economy consisting of S types of men and women. These types are defined by the level of education determined prior to the matching stage. Denote n^{qr} as the number of marriages between type- q men and type- r women, n^{q0} as the number of single type- q men, and n^{0r} as the number of single type- r women. Also, M^q is the number of type- q men, and F^r the number of type- r women. The following accounting identities must hold:

$$n^{q0} + \sum_{r=1}^S n^{qr} = M^q \quad \forall q = 1, \dots, S \quad (12)$$

$$n^{0r} + \sum_{q=1}^S n^{qr} = F^r \quad \forall r = 1, \dots, S \quad (13)$$

In this TU model, a type- q man must transfer π^{qr} amount of utility to a type- r woman to marry her. The utility of type- q man m marrying a type- r woman at time t is

$$V_m^{qr} = \mathbb{E}(v_m^{qr}) - \pi^{qr} + \psi^{qr} + \varepsilon_m^{qr} \quad (14)$$

where $\mathbb{E}(v_m^{qr})$ is the expected economic utility of man m married to a type- r woman, ψ^{qr} is the noneconomic utility ("marital bliss") enjoyed by the couple, and ε_m^{qr} is m 's random preference for the match drawn independently and identically from the type I extreme-value distribution. Equation (14) encompasses the case of singlehood, denoted by $r = 0$, with $v_m^{q0} \equiv \hat{v}_m^q$, $\pi^{q0} = 0$, and $\psi^{q0} = 0$.

Similarly, the utility of type- r woman f marrying a type- q man is

$$V_f^{qr} = \mathbb{E}(v_f^{qr}) + \pi^{qr} + \psi^{qr} + \varepsilon_f^{qr}$$

where ε_f^{qr} is f 's random preference for the match drawn independently and identically from the type I extreme-value distribution.

The marriage market-clearing condition²⁸ gives an equation for the number of matches:

$$\frac{n^{qr}}{\sqrt{n^{q0}n^{0r}}} = \frac{\mathbb{E}(\check{v}^{qr}) - \mathbb{E}(\hat{v}_m^q) - \mathbb{E}(\hat{v}_f^r)}{2} + \Psi^{qr} \quad (15)$$

²⁸See Appendix E.6 for the derivation of the marriage market equilibrium.

where $\Psi^{qr} \equiv \psi^{qr} + \frac{\ln \mu^{qr} - 2 \ln(1 + \mu^{qr})}{2}$. The first term on the right-hand side of equation (15) captures the expected net economic gain from marriage relative to singlehood, including gains from the couple's joint consumption of the public good. The marriage wedge affects equilibrium matching by changing this net economic gain. A higher wedge lowers the gain from marriage by reducing the value of the wife's market work and potentially shifting her labor supply toward home production. Ψ^{qr} signifies the utility from marital bliss and from the intra-household allocation of resources based on Pareto weights.²⁹

3.3 Education

In this section, I describe the women's educational choice problem, without loss of generality. Woman f chooses the education level with the maximum expected utility:

$$\max_{r=1, \dots, S} \{U_F^r - \xi^r\}$$

where

$$U_F^r = \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(\mathbb{E}(v_f^{qr}) + \pi^{qr} + \psi^{qr} \right) \right] - c_F^r \quad (16)$$

Individuals are forward-looking. The expected utility from schooling level r depends on the consequent matching probabilities in the marriage market and the expected utility in each type of match. The costs of schooling level r , on the other hand, consist of the gender-specific direct utility cost c_F^r and idiosyncratic cost ξ^r , drawn independently and identically from the Type I extreme value distribution.

The distribution of the idiosyncratic schooling costs imply that the probability an individual of gender g chooses schooling level s is

$$\mathbb{P} \left(s = \arg \max_{s'=1, \dots, S} \{U_g^{s'} - \xi^{s'}\} \right) = \frac{\exp\{U_g^s\}}{\sum_{s'=1}^S \exp\{U_g^{s'}\}}$$

The maximum likelihood estimator of this probability is $\frac{F^s}{\sum_{s'=1}^S F^{s'}}$ for women and $\frac{M^s}{\sum_{s'=1}^S M^{s'}}$ for men, i.e. the shares of individuals with education level s for each gender.

Thus, the marriage wedge affects equilibrium education choices through two objects in equation (16): the expected utility associated with each match type and, through the marriage-market equilibrium, the probability that a given schooling choice leads to each match type or to singlehood.

²⁹ ψ_{qr} and μ_{qr} are not separately identified. As shown in Appendix G.3, identifying μ_{qr} would require observing equilibrium transfers (π^{qr}) in the marriage market, which are not available in the data. I therefore focus on identifying Ψ_{qr} , which is sufficient for conducting counterfactual analyses.

3.4 Firms

A representative firm produces market output Y^{mkt} . Although the model includes two market goods—the private good and the public good—I assume both are derived from the same market output. The relative price p simply reflects how much more market output is required to produce one unit of the public good compared to the private good. This simplification is innocuous, as the value of p has no impact on equilibrium outcomes related to education, marriage, or labor supply decisions.

In the baseline model, I adopt the simplest setup on the firm's side, assuming a linear production function in male and female effective labor, $\mathcal{M}$ and $\mathcal{F}$:

$$Y^{mkt} = B(\mathcal{M} + \mathcal{F}) \quad (17)$$

The linear production function implies that market wages are independent of aggregate labor supply. I examine the implications of relaxing this assumption in the counterfactual analysis in Section 5—when $\mathcal{M}$ and $\mathcal{F}$ may change—by also considering a CES production function in male and female labor, adopted from [Acemoglu et al. \(2004\)](#):

$$Y^{mkt} = B[(1 - \lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{\frac{1}{r}} \quad (18)$$

Returning to the baseline linear production function, I normalize the labor provided by single males with schooling level 1, market ability 1, and zero children ($\mathcal{K}_0$), i.e. $\bar{w}_M^{10}(\mathcal{K}_0)$, as one unit of effective labor. Then,

$$\begin{aligned} \mathcal{M} &= \sum_{q=1}^S \sum_{r=0}^S \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{w}_M^{qr}(\mathcal{K})}{\bar{w}_M^{10}(\mathcal{K}_0)} \right) \left(P_M^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \\ \mathcal{F} &= \sum_{r=1}^S \sum_{q=0}^S \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{w}_F^{qr}(\mathcal{K})}{\bar{w}_M^{10}(\mathcal{K}_0)} \right) \left(P_F^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \end{aligned}$$

3.5 Aggregate output

Aggregate output is the sum of market output and home production:

$$Y = Y^{mkt} + Y^{home}$$

where

$$Y^{home} = B \sum_{g \in \{M, F\}} \sum_{(q, r)} \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{h}_g^r(\mathcal{K})}{\bar{w}_M^{10}(\mathcal{K}_0)} \right) \left(1 - P_g^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \quad (19)$$

3.6 Equilibrium

An equilibrium in this economy consists of schooling choice q for a man, schooling choice r for a woman, marital transfers π^{qr} , marriage matches (q, r) , public consumption Q , private consumption C_i , labor market participation L_i , total efficient male labor $\mathcal{M}$, total efficient female labor $\mathcal{F}$, market wage, market output Y^{mkt} , total home production Y^{home} , and aggregate output Y , such that

1. Individuals choose the education level that maximizes expected utility, taking as given the probability of different matches and the associated expected utilities.
2. After schooling choices are made, equilibrium marital transfers $\{\pi^{qr}\}$ equate the supply and demand for each marriage match (q, r) .
3. After the matching stage and exogenous determination of children composition, each individual chooses public good consumption Q , private good consumption C_i , and labor supply L_i to maximize their utility. Married individuals maximize equation (1) jointly with their spouse, while singles maximize equation (7) independently.
4. A representative firm hires effective male labor $\mathcal{M}$ and effective female labor $\mathcal{F}$, and pays wage equal to the technology parameter B in equation (17).³⁰
5. Market output Y^{mkt} is given by equation (17), and total home production Y^{home} by (19). Aggregate output Y is given by the sum of Y^{mkt} and Y^{home} .

3.7 Intuition for aggregate output effects of the marriage wedge

In the model, how does aggregate output respond to changes in the marriage wedge? When the wedge on married women's market wages increases, aggregate output can be affected through each of the three stages (in reverse order) of labor supply, marriage, and education.

First, at the labor supply stage, married women's sorting between market work and home production is less aligned with their comparative advantage. This lowers aggregate output.

Second, at the marriage matching stage, higher disutility from wives working in the market makes marriage less attractive, leading some women who would have married to remain single. These newly single women no longer face the marriage wedge, which fosters them to pursue their comparative advantage, thereby increasing output. Another effect is that newly single men and women now receive singles' wages and home productivities. Wage differences between married and single individuals may reflect productivity differences³¹ arising from occupational choices, firm

³⁰See Appendix I.2.1 for details on how I solve for the equilibrium under the CES production function (equation (18)) through fixed-point iteration.

³¹These wage differences may also partly reflect selection or discrimination. As a conservative robustness check, I test the sensitivity of my counterfactual results to assuming that marriage has no causal effect on productivity (Figure 11,

sorting, job mobility, hours, work schedule flexibility, and work effort—factors shaped directly by marital status and indirectly by differential likelihood of having children.³² Whether married men and women are more or less productive in the market and at home than their single counterparts, and how these differences aggregate, is an empirical question. Hence, the net effect on output at the marriage matching stage is a priori ambiguous.

Third, at the educational choice stage, if there is positive assortative matching on education then the lower likelihood of marriage decreases the young women’s marriage-market incentive to invest in education. In addition, if education raises market productivity more than home productivity, the labor-market incentive to get educated depends on how the expected likelihood of working changes. The likelihood of working in the market depends on the balance between i) the lower likelihood of being subject to the marriage wedge as marriage rates fall, and ii) the higher value of the wedge conditional on marriage. How the labor-market returns to education changes is an empirical question, but if education does decrease on net, then aggregate output would also decrease through lower market wages and home productivities.

The overall effect on aggregate output depends on the relative strengths of these mechanisms governed by the underlying parameter values.

4 Data and Parameter Inference

4.1 Data

To simulate the U.S. economy within the model framework, I use the U.S. decennial census (Ruggles et al., 2024), consisting of 1-in-100 nationally representative random sample of individuals. An advantage of the census is that it collects information on all household members so that labor market participation is observed for both spouses in married couples. As the presence of other income-earning household members may influence individual labor decisions, I restrict the sample to household heads and their spouses. I further limit the sample to individuals aged 25 to 54, after education is complete and when individuals are the most economically active.³³

The model is fitted to census data for each decade, assuming that the data reflects the model equilibrium. Calibrating the model separately by decade allows all model parameters to vary flexibly over time. These are: children composition probabilities $\{d^{qr}(\mathcal{K})\}$, the inverse measure of the

case 4) by fully suppressing wage differences between married and single individuals, conditional on gender, education, and children composition.

³²In a lifecycle analysis of U.S. wages, Goldin et al. (2024) find that mothers face a “motherhood penalty,” trading lower pay for employment at more family-friendly firms. They also find that fathers, by contrast, receive a “fatherhood premium,” especially in time-intensive jobs, and interpret this premium as potentially reflecting higher work effort enabled or motivated by marriage and parenthood.

³³Appendix Figure A2 shows that the age range of 25-54 is the most economically active 30-year window.

dispersion of market and home abilities $\{\theta\}$, group market wages $\{\bar{w}_M^{qr}(\mathcal{K}), \bar{w}_F^{qr}(\mathcal{K})\}$ and home productivities $\{\bar{h}_M^{qr}(\mathcal{K}), \bar{h}_F^{qr}(\mathcal{K})\}$, marriage wedges $\{\tau^{qr}(\mathcal{K})\}$, the expected utility enjoyed by a (q, r) match $\{\Psi^{qr}\}$, and the cost of each schooling level $\{c_M^q, c_F^r\}$.

The practical advantage of my model setup is that it is not demanding on the data; the variables required for parameter inference are market wage, labor force participation status, marital status, education, and children. Since all are first jointly available in 1940, I match the model to the decennial census from 1940 to 2010. I exclude the 1950 census, however, because it lacks information on spousal wages. See Appendix F for details on the specific census samples used each year and on the calculation of market wages.³⁴

Section 4.2 below transparently describes the parameter inference procedure, which I implement in sequential steps: starting with an estimate of θ , each subsequent parameter is estimated using the results from earlier steps.

Group construction Some model parameters, such as group market wages or home productivities, are specific to a group, defined by gender, marriage match (i.e., schooling pair), and children composition. As I estimate them by matching population moments to their sample analogs within each group,³⁵ the groups must be sufficiently large for the law of large numbers to apply. At the same time, because individuals within a group share the same parameters, the grouping must also be sufficiently specific.³⁶ Among the grouping variables, schooling underwent the most substantial changes over the sample period. To account for shifting educational distributions over time, I construct five or six schooling categories for each decade, ensuring that at least 5% of the sample falls into each category. For children composition, I similarly define categories to balance group size with group specificity. Since the most pronounced differences in home production duties relate to the presence of a first child and whether the child is young, these factors form the basis for categorization.³⁷

Identification of key parameters The key identification issue is how the model separates the marriage wedge (τ) from group market productivity ($\bar{w}$) and home productivity ($\bar{h}$). For each decade and group, I observe the labor force participation rate and average wage among market workers. Given θ , equation (11) uses these two empirical moments to recover $\bar{w}$. Although participation is endogenous in the model, observed participation can be used in this step because, under the Fréchet assumption, it is a sufficient statistic for selection into market work; recovering $\bar{w}$ does not

³⁴Market wages are computed by dividing total pre-tax wage and salary income from the previous year by the total number of hours worked in that year.

³⁵For example, see equations (10) and (11), applied in steps 3 and 4 of Section 4.2.

³⁶Appendix Table A1 shows that education is by far the strongest predictor of attitudes toward the role of married women, followed by survey year and the number of children—which are all incorporated into the group definitions and decade-by-decade calibration.

³⁷The schooling level and children composition categorizations are in Appendix Tables A2 and A3.

require separately identifying the primitives that generated participation. Home productivity $\bar{h}$ is then identified by the participation rates of men and single women, who do not face the marriage wedge. For women, the key normalization is that home productivity depends on own education and children composition, but not directly on marital status or spouse's education. Thus, conditional on their respective inferred $\bar{w}$ and the $\bar{h}$ normalization, any remaining participation difference between married and otherwise similar single women is assigned to τ .

4.2 Step-by-step parameter inference

I now describe the sequential parameter inference procedure, emphasizing the empirical moment and model equation used at each step.

1. $d^{qr}(\mathcal{K})$: probability of a (q, r) match having children composition $\mathcal{K}$

Set at the empirical probabilities.

2. θ : inverse measure of dispersion of market and home abilities

I estimate θ by maximum likelihood, using the fact that market wages among workers follow a Fréchet distribution. Where x_n is the market ability of observation n and P_n denotes the fraction of workers in observation n 's group, the maximum likelihood estimator for θ is:³⁸

$$\tilde{\theta}_{MLE} = \arg \max_{\theta \in (0, \infty)} \sum_{n=1}^{Nobs} [\ln \theta - \ln P_n - x_n^{-\theta} P_n^{-1} - (\theta + 1) \ln x_n]$$

3. $\bar{w}_g^{qr}(\mathcal{K})$: group market productivity per unit of effective labor

Using the estimate of θ found in step 2 and the average wage and proportion of market-workers in each group in the data, I can back out $\bar{w}_g^{qr}(\mathcal{K})$ from equation (11).

$$\bar{w}_g^{qr}(\mathcal{K}) = \text{avgwage}_g^{qr}(\mathcal{K}) (P_g^{qr}(\mathcal{K}))^{\frac{1}{\theta}} \frac{1}{\Gamma(1 - 1/\theta)}$$

4. $\bar{h}_g^{qr}(\mathcal{K})$: group home productivity per unit of effective labor

$\bar{h}_M^{qr}(\mathcal{K})$ can be backed out from equation (10), armed with θ found in step 2 and the average wage and proportion of market-workers in each group of men in the data. For $\bar{h}_F^r(\mathcal{K})$, I use the average wage and proportion of market-workers among *single* women with r years of schooling and children composition $\mathcal{K}$.

$$\bar{h}_F^r(\mathcal{K}) = \text{avgwage}_F^{0r}(\mathcal{K}) (1 - P_F^{0r}(\mathcal{K}))^{\frac{1}{\theta}} \frac{1}{\Gamma(1 - 1/\theta)}$$

³⁸How I derive the likelihood function and how I isolate market abilities from observed market wages are detailed in Appendixs G.1 and G.2, respectively.

5. $\tau^{qr}(\mathcal{K})$: group marriage wedges

$\tau^{qr}(\mathcal{K})$ is backed out from equation (10) using $\bar{w}_F^{qr}(\mathcal{K})$, $\bar{h}_F^r(\mathcal{K})$, and the fraction of market workers in a group of married women. The idea is that marriage wedges are high if the fraction working in the market is much lower than is predicted from market and home productivities.

$$\tau^{qr}(\mathcal{K}) = 1 - \frac{\text{avgwage}_F^{0r}(\mathcal{K})}{\text{avgwage}_F^{qr}(\mathcal{K})} \left(\frac{1 - P_F^{0r}(\mathcal{K})}{1 - P_F^{qr}(\mathcal{K})} \right)^{\frac{1}{\theta}} \quad (20)$$

6. Ψ^{qr} : utility from marital bliss and intra-household resource allocation, in a (q, r) match

From equation (15),

$$\Psi^{qr} = \frac{n^{qr}}{\sqrt{n^{q0}n^{0r}}} - 2A^{qr} + \hat{A}_M^q + \hat{A}_F^r \quad (21)$$

where

$$A^{qr} = \sum_{\mathcal{K}} d^{qr}(\mathcal{K}) \mathbb{E} \left[\ln \left(\bar{w}_M^{qr}(\mathcal{K}) \epsilon_m^w L_m^* + \bar{h}_M^{qr}(\mathcal{K}) \epsilon_m^h (1 - L_m^*) + \right. \right. \\ \left. \left. [1 - \tau^{qr}(\mathcal{K})] \bar{w}_F^{qr}(\mathcal{K}) \epsilon_f^w L_f^* + \bar{h}_F^{qr}(\mathcal{K}) \epsilon_f^h (1 - L_f^*) \right) \right]$$

and

$$\hat{A}_g^s = \sum_{\mathcal{K}} d^{qr}(\mathcal{K}) \mathbb{E} \left[\ln \left(\bar{w}_g^{qr}(\mathcal{K}) \epsilon_i^w \hat{L}_i^* + \bar{h}_g^{qr}(\mathcal{K}) \epsilon_i^h (1 - \hat{L}_i^*) \right) \right] \quad (q = 0, r = s \text{ if } g = F, \text{ and } q = s, r = 0 \text{ if } g = M)$$

There are no closed-form expressions for A^{qr} or $\hat{A}_g^s$ so I simulate them to back out Ψ^{qr} .³⁹

7. c_g^s : utility cost of acquiring education s

Use equation (16). $\{c_F^r\}_{r=1, \dots, S}$ are found as the solution to the system of equations

$$\frac{F^r}{\sum_{r'=1}^S F^{r'}} = \frac{\exp \left\{ \sum_{q=0}^S \frac{n^{qr}}{F^r} H^{qr} - c_F^r \right\}}{\sum_{r'=1}^S \exp \left\{ \sum_{q=0}^S \frac{n^{qr'}}{F^{r'}} H^{qr'} - c_F^{r'} \right\}} \quad \forall r = 1, \dots, S \quad (22)$$

where

$$H^{qr} = \begin{cases} 2\hat{A}_F^r & \text{if } q = 0 \\ 2A^{qr} + \Psi^{qr} + \frac{\ln n^{q0} - \ln n^{0r}}{2} - \hat{A}_M^q + \hat{A}_F^r & \text{if } q = 1, \dots, S \end{cases}$$

Additionally, $\{c_M^q\}_{q=1, \dots, S}$ are found similarly.⁴⁰

³⁹To permit a more realistic curvature of the utility function, I use the CRRA utility function instead of log utility in simulating A^{qr} and $\hat{A}_g^s$. Note that the CRRA utility fully conforms with the set of conditions of the general utility function, described in Appendix E.1. The coefficient of relative risk aversion used is 0.71, which Chetty (2006) finds to be the average value across 33 different estimation methods spanning structural lifecycle models, natural experiments, earned income responses, and macroeconomic trends.

⁴⁰Detailed derivations are in Appendix G.3.

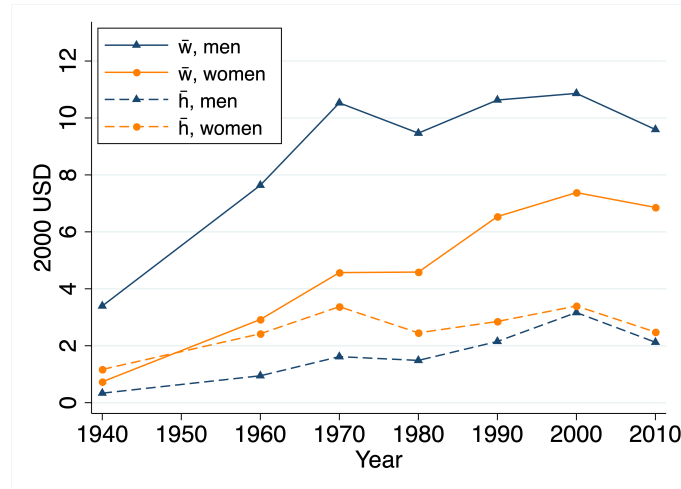
4.3 Calibration results and discussion

In this section, I discuss the calibrated parameter values and how they align with comparable estimates in the literature, related measures from external data sources, or well-established stylized facts. I place particular focus on the calibrated marriage wedges τ .

θ : Dispersion of market and home abilities Appendix Table A4 reports the maximum-likelihood estimates of θ , which range from 1.55 to 1.98 across years. These values are close to Hsieh et al. (2019)’s estimate of 1.52 for the Fréchet shape parameter governing the dispersion of abilities across occupations, as well as their chosen value of 2 for counterfactual analysis. In addition, Appendix Figure A3 shows that the inferred distribution of market abilities closely resembles a Fréchet distribution, lending further support to the assumption of Fréchet-distributed abilities.

$\bar{w}, \bar{h}$: Group market and home productivities Group market wage $\bar{w}$ increases with both the average wage of workers in the group and the group’s labor force participation (LFP) rate. The dependence on the LFP rate reflects selection effects. Higher LFP implies weaker positive selection on idiosyncratic market ability, prompting the model to assign a higher value to $\bar{w}$ to reconcile the observed average wage. In contrast, group home productivity $\bar{h}$ rises with average wage of the workers in that group but decreases in the LFP rate of that group.

Figure 5: Group Market Productivity ($\bar{w}$) and Home Productivity ($\bar{h}$) by Sex



Note: This figure plots the weighted average of the group components of market productivity and home productivity by sex and decade.

Figure 5 plots the averages of $\bar{w}$ and $\bar{h}$ by sex and decade, weighted by the empirical probability of each group. Male market productivity rises steadily from 1940 to 1970 and then plateaus through 2010. Female market productivity, while consistently lower than men’s, shows a more sustained

and pronounced increase, gradually narrowing the gender wage gap. With the exception of women in 1940, home productivity remains lower than market productivity, although it also grows over time. Women’s home productivity is consistently higher than men’s, but the gap narrows over time. These patterns are consistent with rising female attachment to the labor market and persistent—but gradually weakening—gender differences in comparative advantage across market work and home production.

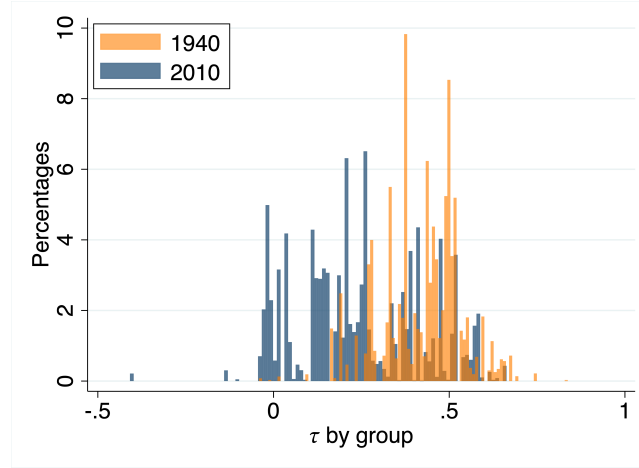
How do $\bar{w}$ and $\bar{h}$ vary by education? As discussed in Section 3.7 on the intuition behind aggregate productivity effects from changing marriage wedges, the relative increase of $\bar{w}$ versus $\bar{h}$ with education is crucial for educational choices in counterfactual scenarios. Specifically, if a young woman anticipates a higher likelihood of market work due to a reduction in the wedge, she will choose to invest more in education only if education raises market productivity more than home productivity. Appendix Table A5 confirms that, for both sexes, education increases market productivity more than home productivity.

τ : Marriage wedges τ represents the wedge on the market wage of married women. To illustrate, a value of $\tau = 0.6$ means that a married woman perceives a \$10 market wage as effectively worth only \$4 when making her labor supply decision.

What exactly does τ encompass? τ is a catch-all term that includes any reason why married women’s LFP differs from that of single women beyond wage differences. It is important to be clear about what factors are—and are not—captured by τ . Because observed wage differences are absorbed into group-specific market productivities $\bar{w}$ before τ is inferred, τ is not confounded by differences in market abilities between married women and similar single women. On the other hand, τ includes many factors that fall under the broad concept of homemaker norms for married women, either self-imposed or expected by others: a) women’s preferences to conform to traditional roles as wives, b) differential valuation of staying home among married women relative to single women,⁴¹ c) differential non-wage treatment of married women by employers relative to single women, and d) men’s preference to have more home-productive women as wives. These channels motivate interpreting τ as closely related to gender norms. At the same time, τ is broader than norms alone. One factor included in τ that does not fall under homemaker norms is that married women make labor supply decisions within two-person households where specialization is possible, whereas single women operate within one-person households. I address this issue by also calibrating the extension with income effects (Appendix E.4) that directly incorporates this difference, discussed further below.

⁴¹See Appendix E.3 for an explicit demonstration.

Figure 6: Histogram of Marriage Wedges (τ)



Note: Histogram of the calibrated $\tau^{qr}(\mathcal{K})$ values for 1940 and 2010. Each group is weighted by its empirical probability.

Figure 6 plots the weighted histograms of τ , calibrated by group, for the years 1940 and 2010. The 2010 distribution clearly sits to the left of the 1940 distribution, indicating a decline in marriage wedges over time.

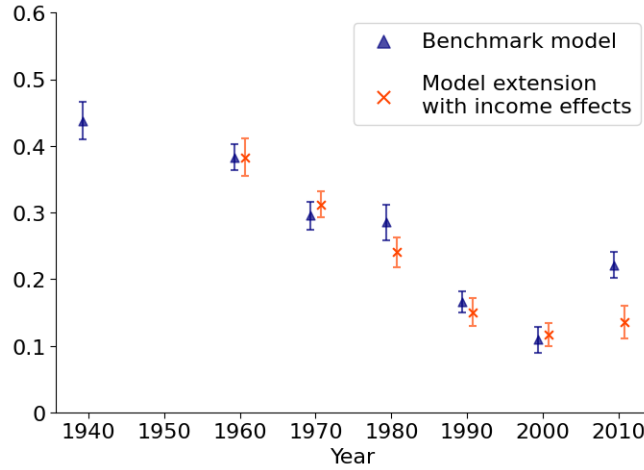
I take the inferred group-specific τ as noisy estimates of marriage wedges, as there are selection effects unaccounted for in the model, such as the correlation between education choice and market abilities, or the correlation between taste for spousal type and market abilities. Therefore, I consider the weighted median of τ by decade.⁴² Figure 7 plots these values in blue. τ generally decreases over time, except from 2000 to 2010. The mathematical explanation for the rise at the end is that, although married women's wages increased relative to those of single women during this period, their LFP did not.⁴³ To reconcile these observations, τ must increase.

Further insight comes from comparing the τ values in the benchmark model to those in the extension with income effects, as shown in Figure 7. In the extension, married women's labor supply decisions respond to their husbands' market earnings; household income effects reduce their incentive to contribute additional market income. Notably, τ barely rises between 2000 and 2010 in the extension, indicating it is the failure to capture spousal dependence that leads the benchmark model to overestimate τ in 2010. Nonetheless, the benchmark model performs well up to 2000.

⁴²The results are very similar when using the weighted average, unweighted average, or unweighted median. I prefer the median, as it is less sensitive to outliers that arise in groups with low empirical probability.

⁴³Figure 1 (or Appendix Figure A8) illustrates that female LFP has plateaued since 1990. The rise in τ persists even when considering market work *hours* of married women relative to single women, indicating that it is not driven by the simple LFP measure failing to capture increases in hours.

Figure 7: Evolution of Marriage Wedges τ



Note: This figure plots the weighted median of τ by decade, in the benchmark model and in the model extension with income effects. The weight of each group corresponds to its empirical probability in the data. The error bars indicate 95% confidence intervals based on bootstrapped standard errors from 100 replications. The τ in the extension can only be estimated from 1960, because household income beyond one's own earnings (i.e., spousal earnings, welfare benefits, social security income, unearned income) are reported only from 1960.

The marriage wedges, though simple to impute, are meaningfully related to gender norms. Table 1 shows that the values of the wedges strongly correlate with survey measures of attitudes about gender roles in marriage. To obtain geographic variation in τ , I redo the parameter inference procedure at the state and state $\times$ decade levels.⁴⁴ I compare these wedges with three attitude measures. First, at the state level, I use a single question⁴⁵ that has been asked the most consistently over time (1938-1998)—whether a married woman should work—and an index averaging responses across all 14 questions shown in Figure 3. Second, at the state $\times$ decade level, I use only the question on whether a married woman should work, so that changes over time are not driven by the changing composition of survey questions. Third, I use individual-level responses to all 14 questions, with question fixed effects. States with higher τ have a greater share of respondents disapproving of married women working (column 1)⁴⁶ and answer more conservatively on the 14 questions combined (column 2). The relationship also holds within states over time: state $\times$ decade τ remains strongly correlated with the state $\times$ decade average response to the single question, even with state

⁴⁴The group categories are reformulated to ensure, as before, that each group is specific enough but also large enough. For τ by state, a group is defined by state, schooling pair, and age cohort: ages 25–34, 35–44, and 45–54. For τ by state $\times$ decade, decade is added to the group definition. I then compute the weighted median of group-level τ for each state or state $\times$ decade.

⁴⁵The single question actually combines responses from two nearly identical questions—[1] and [2] in Appendix Figure A1. See the figure for the exact wordings. Question [1] was asked in the Gallup Polls, and question [2] in the General Social Survey. As shown in the figure, the yearly average trends of the two questions closely overlap, lending confidence that they can be treated as a single question.

⁴⁶Appendix Figure A4 visually shows the tight relationship in column 1.

fixed effects (column 3). Finally, individual-level regressions pooling responses to all 14 questions show that the positive relationship persists, albeit more weakly, both with and without demographic controls (columns 4-5). These robust correlations support the interpretation that gender norms are an important component of the marriage wedge.

Table 1: Correlation between Marriage Wedges and Attitudinal Survey Answers

	State-level attitudes		State×decade-level attitudes	Individual-level attitudes	
	Against married women working	All questions	Against married women working	All questions	
	(1)	(2)	(3)	(4)	(5)
τ by state	0.487*** (0.149)	0.239*** (0.085)			
τ by state×decade			0.237** (0.110)	0.049* (0.028)	0.049* (0.029)
N	51	51	233	188,333	134,661
State FE	-	-	Yes	Yes	Yes
Decade FE	-	-	Yes	Yes	Yes
Question FE	-	-	-	Yes	Yes
Demographic controls	-	-	-	No	Yes

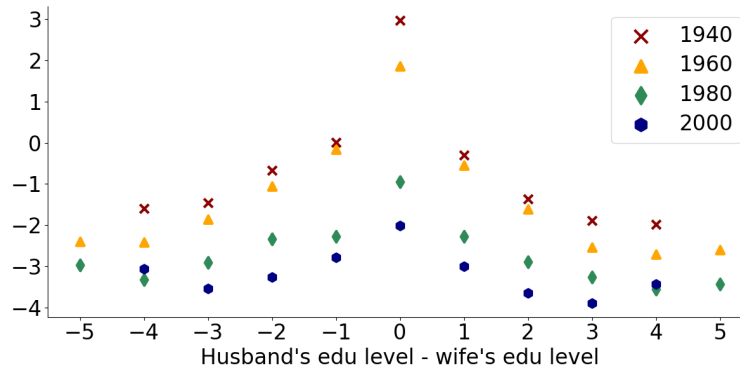
Note: This figure shows the correlation between model-based marriage wedges and survey measures of attitudes. The attitudinal questions (from Figure 3) are coded so that higher values indicate more traditional views. State-level and state×decade-level attitudes are computed as sampling-weighted averages of individual responses. “Against married women working” corresponds to answering “No” to the question: *Should a married woman earn money if she has a husband capable of supporting her?*, asked 1938-1998. Columns 1-2 show cross-sectional regressions at the state level, weighted by state population (robust standard errors). Column 3 reports a state×decade-level regression with state and decade fixed effects, weighted by population (standard errors clustered by state). I do not estimate a state×decade-level regression using all questions because the questions vary in timing, making attitudes aggregated at that level non-comparable over time. Columns 4-5 regress individual responses to the 14 questions on state×decade-level τ , with state, decade, and question fixed effects (standard errors clustered by state). Column 5 adds demographic controls (age, sex, and race dummies) for greater precision, but some questions then drop out due to missing demographic information. * $p < 0.10$, ** $p < 0.05$ *** $p < 0.01$

Finally, Appendix H.2 demonstrates the robustness of τ to sample perturbations and alternative assumptions, including several that address additional differences between married and single women unrelated to norms. This stability provides additional support for interpreting gender norms as a salient part of τ . τ is robust to: 1) incorporating differential tax treatment by marital status by using imputed after-tax wages instead of gross wages; 2) including business and farm income in addition to salary income (from 1960); 3) holding θ fixed, as done in Hsieh et al. (2019), instead of allowing it to vary by decade; 4) dividing married and single individuals by legal marital status

instead of using the IPUMS-constructed “probable spouse” measure, which includes cohabiting couples from 1990; 5) excluding widows from single women to assess sensitivity to the composition of singles; 6) limiting the sample to individuals aged 40–54, rather than 25–54, to account for differences in childbearing potential between married and single women; 7) fixing education levels in all decades (less than high school, high school graduate, more than high school).

Ψ : Utility from marital bliss and intra-household resource allocation Figure 8 plots the average Ψ for each value of the spousal education gap by decade. Within each decade, Ψ exhibits an approximately single-peaked shape, with the peak occurring when spouses have the same education level. Moreover, the decline in the values of Ψ over time indicates a reduction in the overall attractiveness of marriage. These patterns align with the well-documented prevalence of assortative matching by education and falling marriage rates in the U.S. (e.g., Greenwood et al., 2016; Eika et al., 2019).

Figure 8: Ψ by Spousal Education Gap



Note: This figure plots, by decade, the average Ψ^{qr} for each value of the difference in the husband’s and the wife’s education levels ($q - r$), weighted by the empirical probability of each marriage match (q, r).

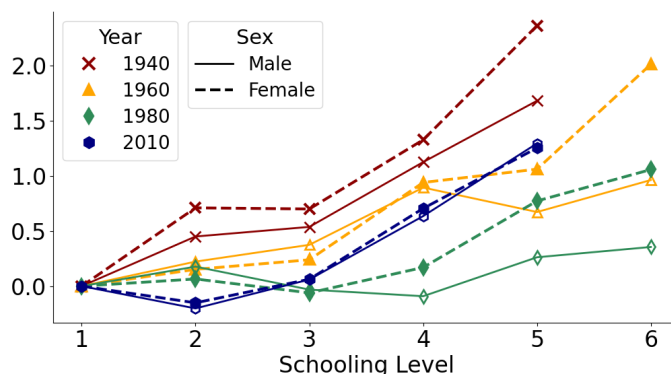
c : Utility cost of schooling The final set of parameters is the gender-specific cost of schooling, c_g^s . These costs are inferred by comparing observed educational attainment to the levels predicted by expected utility, which incorporates both marriage-market returns and economic (wage and home productivity) returns. For example, if more women attain high school than the model predicts based on expected utility, the inferred cost of that level for women must be relatively low.

Figure 9 plots the values of c_g^s in selected years, divided by the average economic utility (A^{qr}) of that year for greater comparability over time.⁴⁷ It shows that the estimated cost of schooling c increases with education level in all years. While changes in education categories across decades limit direct comparisons over time, broad patterns emerge. Initially, the costs decline—consistent

⁴⁷Raw values of c_g^s for every decade are reported in Appendix Table A6.

with rising educational attainment—but rise again in later decades, potentially reflecting increasing pecuniary costs of post-secondary education. Within each decade, c values are comparable across sex. Early cohorts show higher costs for women, especially at advanced education levels, but the gender gap narrows over time. By 2010, male and female cost profiles are nearly identical, and the male cost of attaining the highest schooling level even exceeds the female cost. This pattern reflects the well-documented reversal in gender gaps in educational attainment in the U.S., where women began surpassing men in college completion rates from the mid-1990s.

Figure 9: Cost of Schooling c by Sex and Decade



Note: This figure plots, by decade, the utility cost of schooling (c_g^s) normalized by the average economic utility (A^{qr}) for that decade. c_g^1 is normalized to zero for each year and gender. The categorization of schooling levels, which varies by decade, is listed in Appendix Table A2.

5 Counterfactual Exercises

To quantify the aggregate importance of the marriage wedge, I study how aggregate output Y responds when τ changes while all other parameters are held fixed. The steps for conducting counterfactuals are described in Appendix I.1.

Main Counterfactual Table 2 shows how various aggregate variables in 2000—the latest decade in which the benchmark model performs well—would change if the marriage wedge had remained at its 1940 level, holding all other parameters fixed at 2000 values. Because group-specific marriage wedges are likely noisy estimates, I use the weighted median τ for each decade (plotted in Figure 7). Applying the same value of τ to all individuals in the counterfactuals also ensures that changes in group sizes across scenarios do not mechanically alter the mean or distribution of τ in the economy.

I consider two adjustment margins: when only labor supply responds (column 1), and when education, marriage, and labor supply all respond (column 2).

**Table 2: Percentage Changes in Various Aggregate Variables
if Individuals of 2000 were Subject to the Marriage Wedge of 1940**

	Adjustment margins: Labor supply (1)	Education, marriage, labor supply (2)
Education		
Women's average years of schooling	-	-2.6
Men's average years of schooling	-	-2.0
Selection into marriage		
Marriage rate	-	-26.3
Married women's average years of schooling	-	-5.3
Single women's average years of schooling	-	1.2
Married men's average years of schooling	-	-4.3
Single men's average years of schooling	-	0.9
Latent market and home productivity		
Women's average $\bar{w}$	-	-2.9
Men's average $\bar{w}$	-	-7.2
Women's average $\bar{h}$	-	-0.4
Men's average $\bar{h}$	-	3.3
Labor Force Participation		
Married women's LFP	-24.8	-30.8
Single women's LFP	-	0.8
Married men's LFP	-	-1.6
Single men's LFP	-	0.4
Output per head		
Married women's market output	-12.3	-23.8
Married women's total output	-2.9	-10.6
Married men's market output	-	-7.6
Married men's total output	-	-6.2
Aggregate market output	-3.6	-8.7
Aggregate total output	-0.9	-5.3

Note: This table reports the percentage changes in various aggregate variables that occur when the individuals of 2000 are subject to the marriage wedge of 1940, holding all other parameter values constant at the 2000 level. Column 1 holds the marriage match patterns and educational choices constant at the 2000 level and considers only changes to the married individuals' labor supply decisions. Column 2 additionally allows the (forward-looking) marriage matching and educational choices to change in accordance with the new expected utilities arising from the altered labor supply behavior.

Column 1 isolates the direct effect of the marriage wedge by holding marriage and education choices fixed. When τ rises from 0.11 in 2000 to 0.44 in 1940, married women's labor force participation drops by 24.8%. This reduces their cumulative market output Y^{mkt} by 12.3%. At the same time, home production Y^{home} increases, offsetting part of the loss. As a result, their

cumulative total output $Y = Y^{mkt} + Y^{home}$ declines by only 2.9%. In the aggregate, factoring in the share of married women in the population, market output falls by 3.6% and total output by 0.9%. The dissimilar effects on Y^{mkt} and Y highlight the importance of accounting for nonmarket output, often excluded from national accounts. The output gains when women enter the labor market would be overstated in methods that only consider market output.

In column 2, higher τ not only affects labor supply directly but also indirectly alters education and marriage decisions. Since the marriage wedge is modeled as marriage-specific costs, a higher τ makes marriage less attractive, leading to a 26.3% drop in the marriage rate. As the marriage wedge is more costly for women with higher market wages, higher-educated women are the first to drop out of marriage. Due to assortative marriage matching by education, the fall in marriage is more pronounced also for higher educated men. As the set of married women is now more negatively selected on education, married women's LFP rate and cumulative market output fall by more in column 2 than in column 1. Married women's cumulative total output falls by 10.6%.

Furthermore, average years of schooling decline for both men (by -2.0%) and women (-2.6%), with a more pronounced drop among women.⁴⁸ For men, this decline is driven solely by reduced returns in the marriage market. For women, education decisions respond to changes in both marriage- and labor-market returns. The direction of change in labor-market returns is ex ante ambiguous. Highly educated women, now more likely to remain single, avoid the marriage wedge and are more likely to participate in the labor market. However, among those who marry, the stronger wedge reduces their likelihood of market work. In this counterfactual, women's labor-market returns to education appear to decline as well, contributing to a larger reduction in educational attainment among women relative to men.

Lower education and marriage rates alter men's and women's latent productivity in the market and at home, captured by the average of $\bar{w}_g^{qr}(\mathcal{K})$ and $\bar{h}_g^{qr}(\mathcal{K})$ by gender. These changes operate through shifts in the size of groups with different market and home productivities. Interestingly, although women's education declines more than men's, their market productivity falls by less (-2.9% for women vs. -7.2% for men). This is because married men have higher $\bar{w}$ than single men, whereas the opposite holds for women.⁴⁹ Thus, for women, lower marriage rates partially offset the productivity loss from lower education; for men, both factors reduce market productivity. Home productivity decreases slightly for men and increases slightly for women, consistent with its modest sensitivity to education (Appendix Table A5) and the fact that single men have higher home productivity than married men.

⁴⁸The decline is even starker when measured by the percentage change in the share attaining the highest education level (postgraduate studies): -10.7% for men, and -14.1% for women.

⁴⁹This pattern aligns with Goldin et al. (2024), who document a U.S. fatherhood wage premium and interpret it as potentially reflecting men's higher work effort, enabled or motivated by marriage and parenthood. They also document a motherhood penalty, driven primarily by lower work hours and sorting into family-friendly firms with lower pay.

Overall, aggregate market output falls by 8.7% and total output by 5.3%. Comparing the output counterfactuals in columns 1 and 2 highlights the importance of accounting for general equilibrium effects. The difference is striking: allowing for endogenous responses in education and marriage markets nearly sixfolds the total output loss from 0.9% to 5.3%. The comparison demonstrates that partial-equilibrium analyses of gendered family norms and institutions, as is often the case in regression-based empirical research, may severely underestimate their full economic impact.

Is the effect of an 8.7% fall in aggregate market output and a 5.3% fall in aggregate total output small or large? Table 3 shows that these drops in aggregate output represent 12.6% of the growth in market output and 8.1% of the growth in total output between 1940 and 2000. The effects might be viewed as large because the counterfactual changes only a narrow object: the wedge faced by married women, after accounting for their wages, children, education, and selection into marriage. It does not include broader narrowing of gender gaps between men and women, or broader changes common to all women. At the same time, the effects are modest relative to the full scale of U.S. output growth over this period. To benchmark the magnitude of these output effects from another angle, I conduct some additional counterfactuals below.

Table 3: Share of Growth Accounted for by the Marriage Wedge

	Actual 1940 (1)	Actual 2000 (2)	Counterfactual 2000 with 1940 τ (3)	1940-2000 growth accounted for by τ (4)
Aggregate market output (2000 USD)	4.7	15.1	13.8	12.6%
Aggregate total output (2000 USD)	6.2	17.5	16.6	8.1%
Share of women with >12 yrs of schooling (%)	10.0	52.8	45.4	17.4%
Share of men with >12 yrs of schooling (%)	11.7	53.1	47.4	13.7%
Share of married women with >12 yrs of schooling (%)	9.9	52.6	36.3	38.3%
Married women's LFP (%)	14.4	74.6	51.7	38.1%
Wives' share of household market income (%)	6.3	36.3	27.3	30.0%

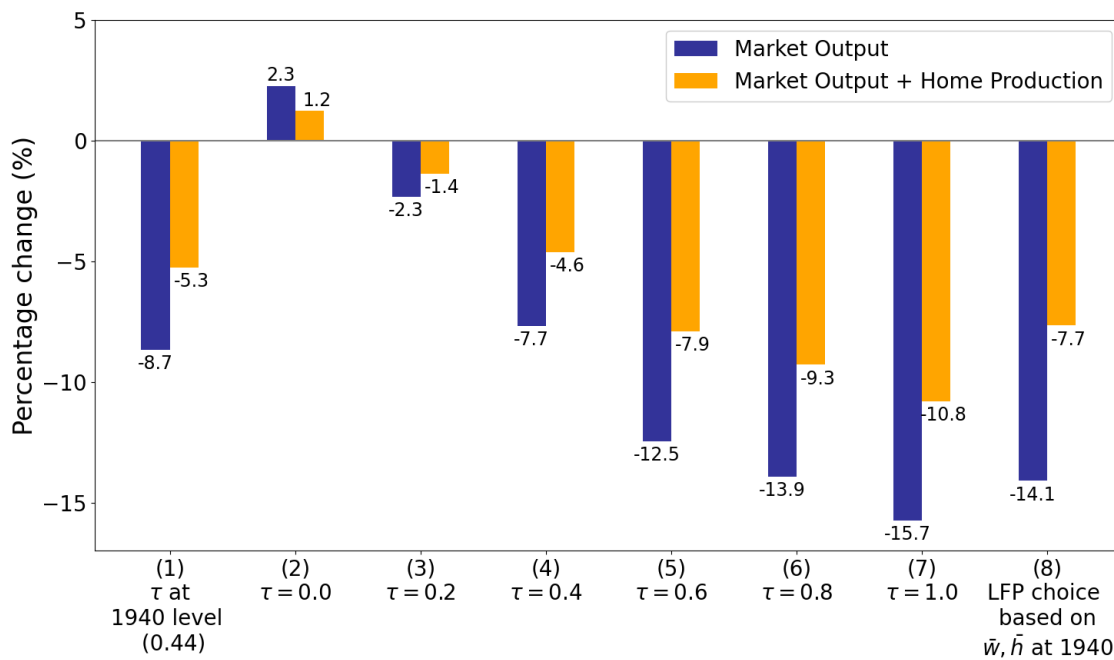
Note: Columns 1-3 report the levels of various aggregate variables in 1940, 2000, and in the counterfactual 2000 economy with the 1940 marriage wedge. Column 4 shows the share of growth attributed to the change in τ between 1940 and 2000. Aggregate market output is normalized to equal the average real hourly wage, while aggregate total output is normalized to equal the sum of the average real hourly wage and the average real hourly home productivity. Wife's share of household market income is her real hourly wage divided by the couple's total wage, with nonparticipation counted as zero. These hourly measures reflect the model's binary labor supply choice—market work or home production—excluding the role for hours worked.

Besides aggregate productivity growth, the decline in the marriage wedge between 1940 and 2000 helps explain several key changes in U.S. economic and social outcomes over this period. As shown in Table 3, the decline in the wedge accounts for 38.1% of the rise in married women's labor

force participation and 30.0% of the increase in wives' share of household market income. It also contributed to growing educational attainment for both men and women, but since it accounts for a larger share of women's gains (17.4%) than men's (13.7%), it helps explain why women's education grew faster than men's during this period. Moreover, it accounts for 38.3% of the increase in married women's education—a substantially larger share than for all women's education. Thus, the decline in τ also helps explain the increasingly positive selection into marriage by education.

Additional Counterfactuals Figure 10 compares the market and total (market and nonmarket) output effects of various counterfactual scenarios, where the baseline year is set at 2000. Besides the main counterfactual (scenario 1), the others explore the effects of τ changing from 0 to 1 (scenarios 2-7), and when labor supply choices of females are made based on $\bar{w}$ and $\bar{h}$ of 1940 (scenario 8). All the three margins of education, marriage, and labor supply are allowed to adjust.

Figure 10: Market and Total Output Effects of Different Counterfactuals



Note: This figure compares the market and total (market and nonmarket) output effects of various counterfactual scenarios, where the baseline year is set at 2000, with τ at 2000 being 0.15.

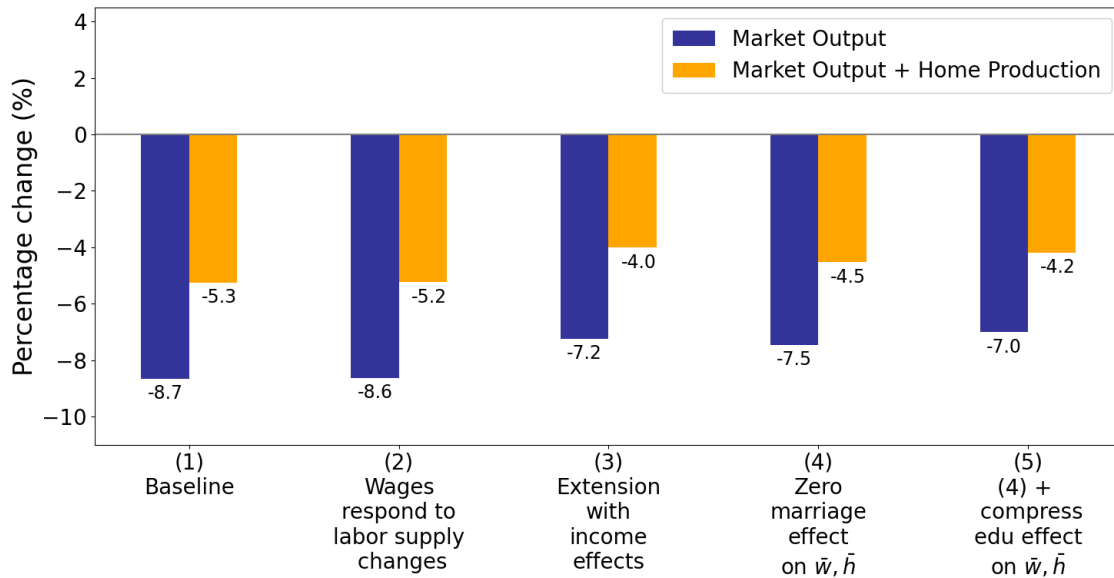
As τ increases, the negative output effects become increasingly pronounced. When married women are completely barred from working ($\tau = 1$), market output falls by 15.7% and total output by 10.8%. This extreme counterfactual illustrates the output stakes in settings where norms or institutions strongly discourage married women's market work. It also shows what may be at stake

if gender attitudes stall or reverse.

I focus on counterfactual scenario 8 to benchmark the effects of the main counterfactual. As illustrated in Figure 5, market and home productivities changed substantially between 1940 and 2000, fueling economic growth over this period. Notably, women's $\bar{w}$ was lower than their $\bar{h}$ in 1940, whereas by 2000, $\bar{w}$ was more than twice as large as $\bar{h}$. To avoid capturing the direct effects of productivity changes on output, this counterfactual distorts female labor supply choices to be based on their 1940 levels of $\bar{w}$ and $\bar{h}$, while actual wages and productivities are held at their 2000 levels. The resulting shift in labor supply is dramatic: the labor force participation rate of married women drops by 67%—over twice as much as in the main counterfactual. As a result, market output falls by 14.1%, and total output declines by 7.7%. The main counterfactual's 5.3% total-output loss is nearly 70% as large as the 7.7% loss in scenario 8, indicating that the marriage wedge has sizable output consequences even relative to large historical advances in women's economic incentives to work in the market.

Extensions and Robustness Figure 11 presents robustness checks and evaluates model extensions for the main counterfactual: the change in output in the 2000 economy with the 1940 marriage wedge imposed.

Figure 11: Extensions and Robustness of the Main Counterfactual



Note: This figure compares the market and total (market and nonmarket) output effects of various counterfactual scenarios, where the baseline year is set at 2000, with τ at 2000 being 0.15.

In case 2, wages adjust to changes in male and female labor supply under a CES production

function where the two types of labor are imperfectly substitutable inputs.⁵⁰ The output effects remain close to the baseline because the estimated wage responses—a rise in female wages and a fall in male wages—are small.

Case 3 implements the counterfactual in the model extension with income effects.⁵¹ The results show smaller output effects than the baseline. This attenuation occurs because the change in expected economic utility within marriage decreases by less under this specification, which induces a smaller decline in marriage rates (-15%, compared to -26% in the baseline) and consequently a smaller reduction in education investment.

Cases 4 and 5 relax the baseline assumption that the observed wage differentials across groups—defined by sex, own education, spousal education if married, and children composition—reflect purely causal effects of belonging in that group. In the baseline, for example, an individual who marries is assumed to experience a change in market productivity equal to the observed difference in $\bar{w}$ between married and single individuals. However, the difference in $\bar{w}$ may also capture selection effects missing from the baseline model, such as sorting into marriage or education based on unobserved ability. If wage differentials are only partly causal, changes in productivity associated with changes in marital status or schooling in the baseline counterfactual may be overstated. To address this issue, I compress wage differentials based on standard assumptions or empirical decompositions from the literature that separate causal effects from selection. This provides a shortcut approach to netting out unmodeled selection effects, without explicitly modeling them. Details on the wage compression in cases 4 and 5 are provided in Appendix I.2.3.

In case 4, I assume a flat wage differential across marital status and partner type, adopting a standard earnings equation specification (e.g., [Chiappori et al., 2018](#)) where earnings do not depend on these factors. This is a conservative specification that completely rules out the causal effect of marriage on productivity.⁵² In light of the child penalty literature pioneered by [Kleven et al. \(2019\)](#), I still allow wages to differ by children composition index—reflecting skill depreciation when out of work, as in [Adda et al. \(2017\)](#), or reduced time availability and flexibility, as in [Goldin \(2014\)](#), due to children. The output effects are smaller than in the baseline, indicating that part of the decline in output in the baseline stems from the fact that the fall in marriage induces more individuals to receive single wages rather than married wages. Specifically, while single women’s $\bar{w}$ are found to be higher than married women’s $\bar{w}$ on average in 2000, the opposite is true for men. Thus, some of the baseline effect comes from more men receiving the lower single wages, which partially offsets

⁵⁰The elasticity of substitution between male and female labor is set to 3, following the estimation result in [Acemoglu et al. \(2004\)](#). Appendix I.2.1 describes how I solve for the equilibrium through fixed-point iteration.

⁵¹See Appendix I.2.2 for details.

⁵²[Korenman and Neumark \(1991\)](#) find that only a small share of the marital wage premium of men is driven by selection on wages, wage growth, or other wage-enhancing characteristics, and instead provide support for the view that marriage itself enhances men’s labor market productivity.

the productivity gains from more women receiving the higher single wages. This component of the output effect is eliminated in case 4 by construction.

In case 5, I further compress the wage differential by education to 85% of its observed level, in addition to the wage compression in case 4. The 85% compression represents a conservative benchmark, since calculations based on estimated education and ability premiums in [Chiappori et al. \(2018\)](#) show that the causal education premium typically accounts for a much larger share of the total wage gap by education.⁵³ As expected, the output effects are even smaller than in case 4, as the lower education investment in the counterfactual now translates to lower reductions in productivity.

Overall, the magnitude of market output loss ranges from -7.0% to -8.7% , while total output loss ranges from -4.0% to -5.3% . Even the smallest loss is roughly half the size of the output loss from labor supply distortions driven by economic forces (scenario 8, Figure 10). The qualitative finding of a substantial decline in both market and total output remains robust to alternative model assumptions.

6 Conclusion

I measure a marriage wedge on married women's market work and study its effects on aggregate output using historical data from the United States. Over the past 70 years, this wedge has weakened substantially, with the implied tax on married women's market wages falling from 44% in 1940 to 14% in 2010. The wedge is strongly related to survey measures of gender-role attitudes, supporting the interpretation that traditional gender norms are an important component.

The marriage wedge has sizable output consequences. Had the 1940 wedge persisted in the 2000 economy, aggregate market output would have been 8.7% lower and aggregate total output 5.3% lower. This amounts to nearly 70% of the output cost of the distortion in female labor supply from *economic* forces—when female labor force participation is based on wages and home productivities of 1940. The decline in the marriage wedge also accounts for 12.6% of the growth in market output and 8.1% of the growth in total output between 1940 and 2000. The effects are sizable given that the counterfactual isolates a marriage-specific margin distinct from the gender-wide changes affecting women regardless of marital status, which are the focus of much of the existing literature.

The magnitude of the output effects reflects the importance of endogenous adjustments in education and marriage markets, as counterfactual effects holding them fixed are significantly more modest. The amplification in general equilibrium occurs through multiple channels. First, a higher

⁵³Appendix J shows my calculations. Moreover, the empirical literature on returns to schooling, as surveyed in [Card \(1995\)](#), typically finds that instrumental variables (IV) estimates exceed ordinary least squares (OLS) estimates. This suggests that the education premium may be larger than observed wage differentials by education—in which case the decline in output would be even larger than in the baseline.

marriage wedge lowers the marriage rate, especially among highly educated women. Given strong assortative matching by education, this weakens the marriage-market incentives to invest in education for both men and women. Second, for women, the reduced likelihood of labor force participation after marriage diminishes the labor-market returns to education, even though fewer women marry. Third, changes in marriage and fertility patterns alter individuals' latent productivities.

My findings suggest that analyses of gendered family institutions that focus solely on immediate labor supply responses may substantially understate their full economic impact, by missing the feedback effects through human capital investment and family formation decisions. For instance, short-term evaluations of policies or social movements that shift norms, or perceptions of norms, may not capture the full impact that would emerge in the long term.

Despite its decline since 1940, the marriage wedge remains well above zero, leaving scope for further output growth and gender convergence if marriage-specific barriers to women's market work continue to recede. Yet its stagnation in recent decades suggests that other margins—such as greater temporal flexibility in work arrangements (Goldin, 2014)—may be needed to help married women combine market work with family responsibilities and reduce the remaining wedge.

The tractability of the model and its transparent parameter-inference procedure offer practical advantages. For instance, Lee (2021) takes the model to 17 other countries and augments it with a male-female wedge, allowing the within-women marriage wedge central to this paper to be compared with broader differences between men and women. At the same time, the model abstracts from important margins such as occupations, divorce, childcare costs and transfers (Hannusch, 2024), and leisure (Aguiar and Hurst, 2007b). Incorporating these margins would allow a richer understanding of how marriage shapes women's labor supply and aggregate output.

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Online Appendix

Appendix A Additional Tables

Table A1: Variation in Attitudes by Individual Characteristics

Against married women working					
	Average	Shapley decomp (%)		Average	Shapley decomp (%)
	(1)	(2)		(3)	(4)
Education			Race		
None or grades 1-8	0.40		White	0.17	
Grades 9-11	0.28		Black	0.24	4.0
Grade 12	0.20	58.5	Other	0.21	
Some college	0.14				
College graduate or higher	0.09				
Survey year			Marital status		
1970-1979	0.25		Married	0.19	
1980-1989	0.16	12.5	Widowed	0.28	
1990-1999	0.16		Divorced	0.17	2.3
			Separated	0.19	
			Never married	0.15	
Number of children			Geographic region		
0	0.14		Northeast	0.17	
1	0.17		Midwest	0.18	1.9
2	0.18	10.1	South	0.21	
3	0.20		West	0.16	
4 or more	0.26				
Age			Sex		
20-29	0.15		Male	0.19	0.7
30-39	0.16	6.5	Female	0.18	
40-49	0.19				
50-59	0.24				

Note: This table reports how much different individual characteristics account for the variation in attitudes about married women working in the labor market. “Against married women working” corresponds to answering “Disapprove” to the question: *Do you approve or disapprove of a married woman earning money in business or industry if she has a husband capable of supporting her?* from the General Social Survey. Columns 1 and 3 report the weighted-average disapproval rate by category of each variable, illustrating the variation across categories. For columns 2 and 4, I regress the disapproval indicator on the dummies for the categories of each variable and report the Shorrocks-Shapely decomposition denoting the relative contribution of each variable to the R-squared.

Table A2: Years of Completed Schooling by Schooling Category by Year

Schooling level	Year						
	1940	1960	1970	1980	1990	2000	2010
1	[0,7]	[0,7]	[0,8]	[0,9]	[0,11]	[0,11]	[0,11]
2	8	8	[9,11]	[10,11]	12	12	12
3	[9,11]	[9,11]	12	12	[13,15]	[13,15]	[13,15]
4	12	12	[13,15]	[13,15]	16	16	16
5	[13,∞)	[13,15]	[16,∞)	16	[17,∞)	[17,∞)	[17,∞)
6		[16,∞)		[17,∞)			

Note: Years of completed schooling are integers in each interval. For example, individuals with 0,1,...,7 years of completed schooling fall under schooling level 1 in 1940.

Table A3: Description of Children Composition Categories

Children composition categories		Description
1		No child
2		1 child, aged 6-18
3		1 child, aged 0-5
4		2 or more children, all aged 6-18
5		2 or more children, at least one aged 0-5

Note: All children are one's own children living in the same household.

Table A4: Maximum Likelihood Estimate of θ

	1940	1960	1970	1980	1990	2000	2010
$\hat{\theta}$	1.55*** (18.2)	1.83*** (28.3)	1.84*** (19.4)	1.72*** (12.6)	1.98*** (22.0)	1.98*** (21.9)	1.84*** (39.4)
N	156,404	315,190	315,424	562,479	699,598	796,976	736,666

Note: t statistics using standard errors clustered by sex in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

When the standard errors are not clustered, following the model assumption of independently drawn market abilities, the t statistic is incredibly large due to the large sample size.

See step 2 of Section 4.2 for the maximum likelihood estimation strategy.

Table A5: Coefficients from Regressing $\bar{w}$ or $\bar{h}$ on Schooling Level

	Men		Women	
	$\bar{w}$	$\bar{h}$	$\bar{w}$	$\bar{h}$
Schooling level	1.555*** (0.373)	-0.017 (0.016)	1.340** (0.366)	0.039 (0.034)
Number of groups	1170	1170	1170	1170

Note: This table reports the coefficients on schooling level from regressions of group market wage ($\bar{w}$) and home productivity ($\bar{h}$) on schooling level, controlling for year, children composition, marital status, and spousal schooling level (all as dummy variables). Each observation corresponds to a group and is weighted by its empirical probability. As described in Appendix Table A2, schooling level is a categorical variable ordered by increasing years of completed schooling. For simplicity, it is treated as continuous in these regressions to illustrate how $\bar{w}$ and $\bar{h}$ vary with education. Standard errors (in parentheses) are clustered by census year to account for within-decade correlation due to decade-by-decade parameter inference and changing categorization of schooling level across census years. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

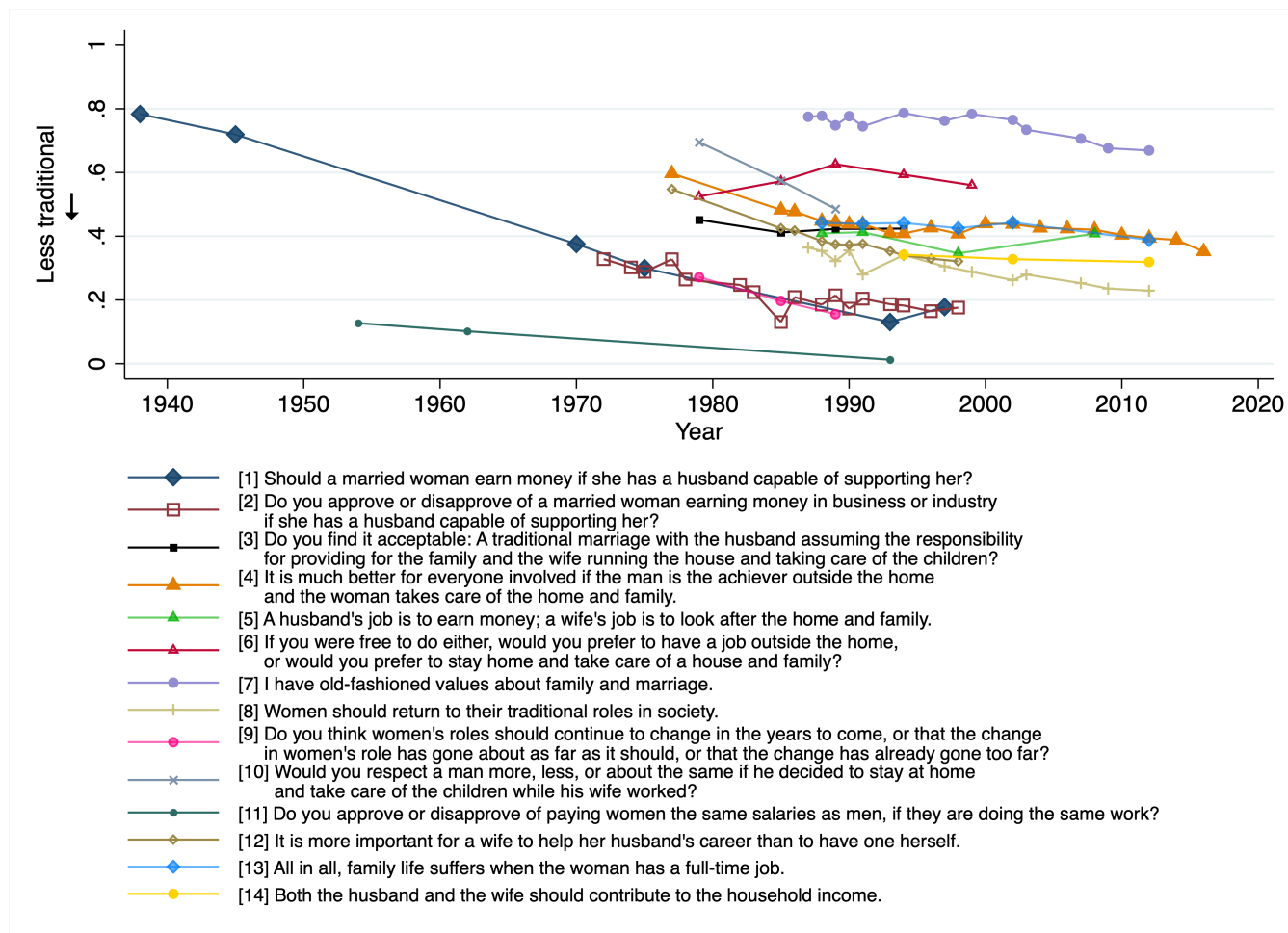
Table A6: Estimates of the Costs of Schooling c by Sex and Decade

		Schooling level					
		1	2	3	4	5	6
1940	Male	0.0	0.67	0.80	1.68	2.51	
	Female	0.0	1.06	1.05	1.98	3.53	
1960	Male	0.0	0.51	0.86	2.04	1.53	2.20
	Female	0.0	0.35	0.55	2.15	2.42	4.59
1970	Male	0.0	0.5	1.25	1.19	2.16	
	Female	0.0	0.02	1.17	2.00	4.60	
1980	Male	0.0	0.49	-0.09	-0.25	0.72	0.98
	Female	0.0	0.18	-0.17	0.47	2.13	2.91
1990	Male	0.0	-0.32	-0.22	0.93	2.51	
	Female	0.0	0.11	0.48	2.33	3.91	
2000	Male	0.0	-0.64	0.10	1.40	3.11	
	Female	0.0	-0.62	0.12	1.71	3.29	
2010	Male	0.0	-0.59	0.18	1.85	3.78	
	Female	0.0	-0.44	0.18	2.06	3.66	

Note: This table reports the estimates of c_g^s , the utility cost of schooling by sex and decade. As a normalization, I set $c_g^1 = 0$. The schooling categories for each decade are described in Appendix Table A2.

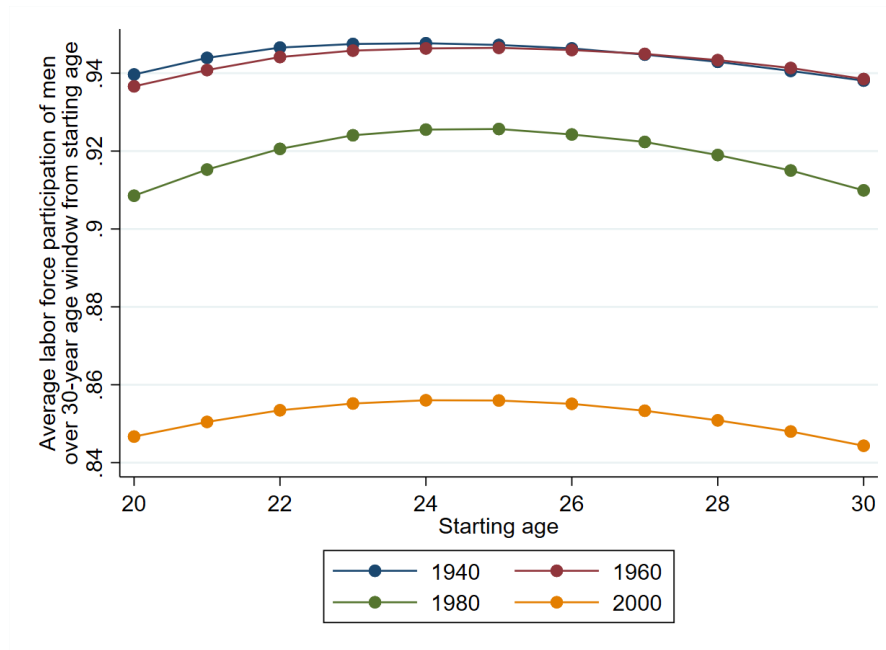
Appendix B Additional Figures

Figure A1: Trends of Attitudes Relating to Gender Roles within Marriage



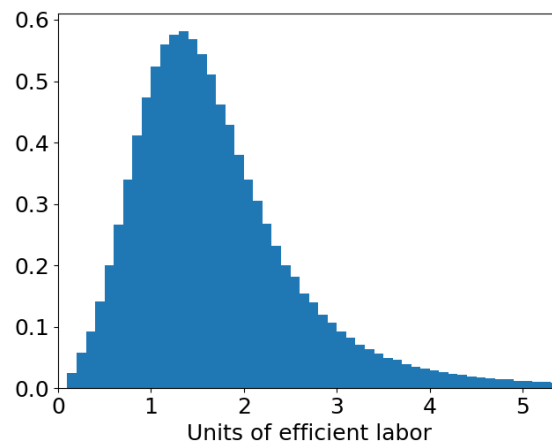
Note: The attitudinal survey data originates from the ROPER polls database and the General Social Survey. For ROPER polls, I perform a keyword search of “women” and “gender,” select the questions relating to gender roles within marriage, and among them, select the ones asked repeatedly over multiple periods. All samples are nationally representative adults or nationally representative men and women separately. All survey answers have been reconfigured to lie between 0 and 1, with 1 corresponding to the most traditional answer and 0 the least. This figure plots the weighted averages of the reconfigured answers by year.

**Figure A2: Average Labor Force Participation of Men
over a 30-Year Window from Starting Age**



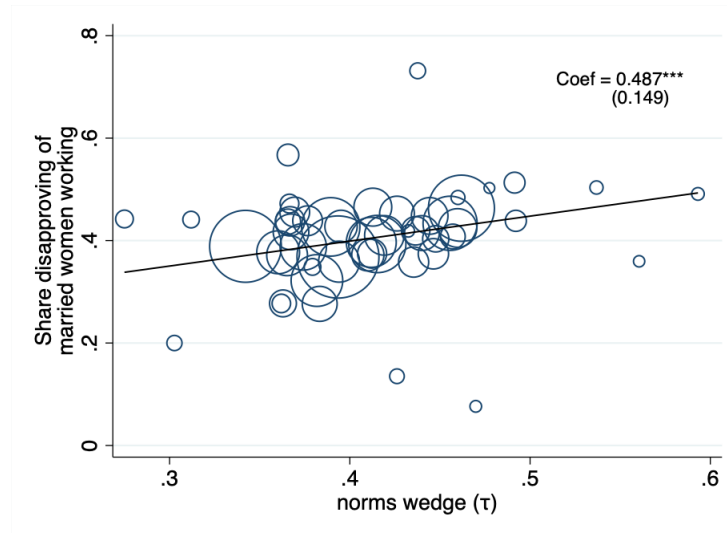
Note: This figure plots the weighted average of labor force participation among men aged between starting age and (starting age+29). It shows that 25-54 is an appropriate age range for the economically active years of one's life and that this observation is quite stable over time.

Figure A3: Histogram of Empirical Market Abilities



Note: This figure plots the histogram of the market abilities of all working individuals with wage data, where the market abilities are inferred using the model structure as outlined in step 2 of Section 4.2.

Figure A4: Conservative States Have Larger Marriage Wedges



Note: This figure visually shows the result in column 1 of Table 1: states with higher τ are also the ones where a greater share of individuals disapprove of married women working. Each observation is a state, and the bubble size represents the state population size.

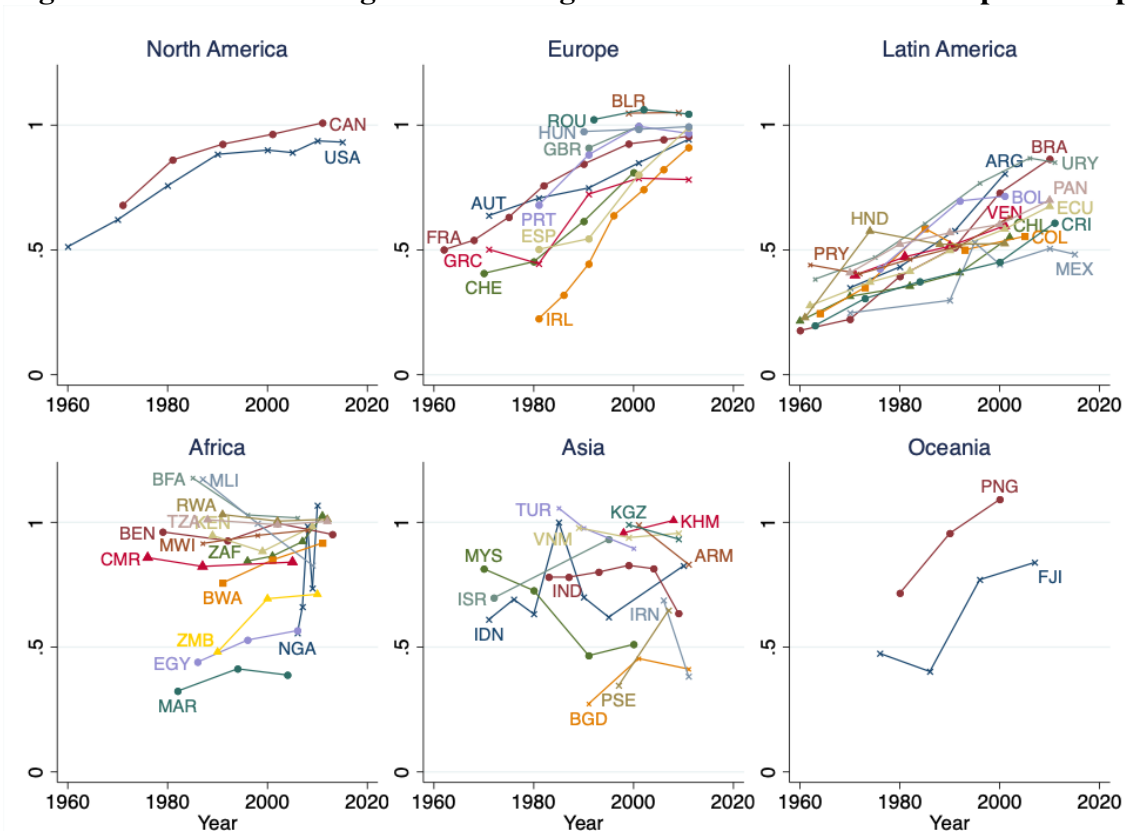
Appendix C Cross-Country Patterns: Married Female Labor Force Participation and Gender Attitudes

In many countries, married women are closing the labor force participation gap with single women, alongside a shift toward less conservative gender attitudes. Wider gaps persist where conservative views are stronger, suggesting that the margin studied in this paper remains relevant beyond the historical United States.

C.1 Married-Single Female Labor Force Participation Ratio

I use the country censuses available on IPUMS International ([Minnesota Population Center, 2019](#)) to compare the labor force participation of married women with that of single women.

Figure A5: The Narrowing Married–Single Female Labor Force Participation Gap



Notes: This figure plots the ratio of the labor force participation rate of married women to that of single women (never married, separated, divorced, or widowed), ages 25-54, by country and year.

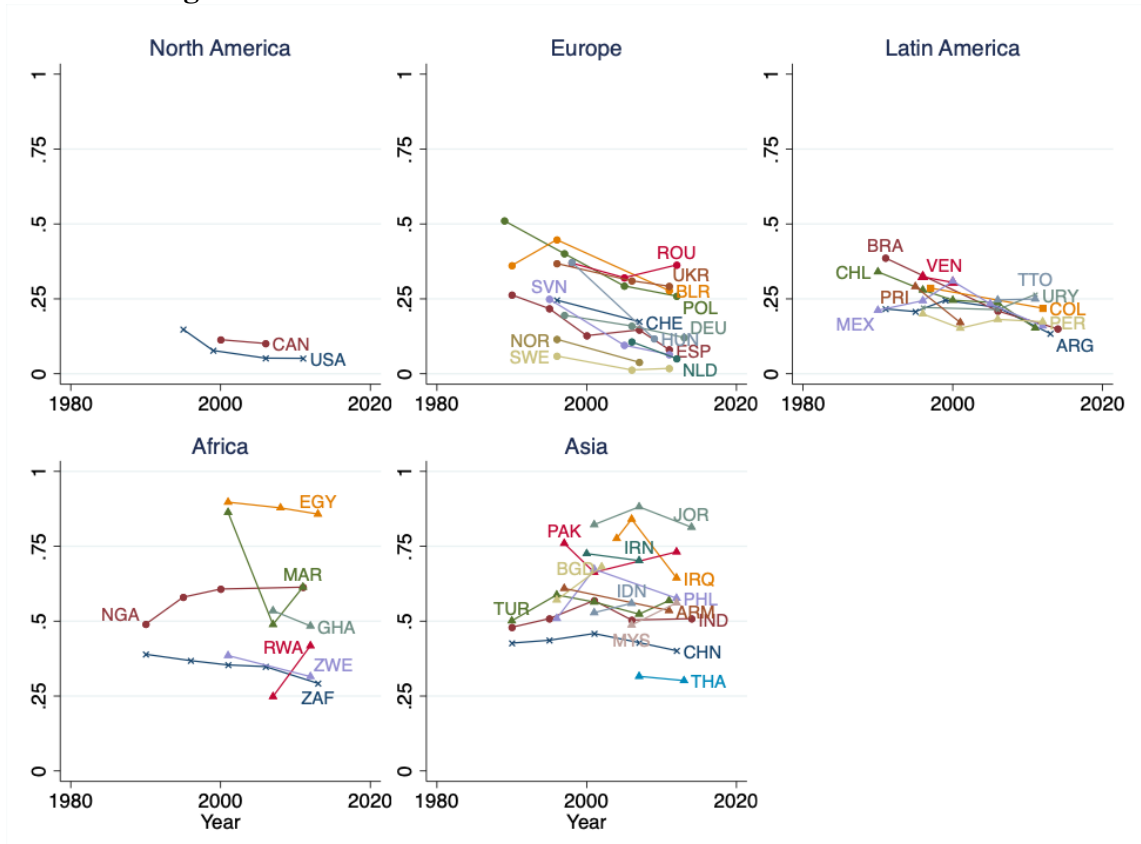
Figure A5 shows that the married-single FLFP ratio has been rising in many parts of the globe, resonating the trend of the U.S. in Figure 1. These not only include the richer OECD countries of

North America and Western and Southern Europe, but also the middle- and low-income countries of Eastern Europe, Latin America, and Oceania. The patterns are more mixed in Africa and Asia.

C.2 Attitudes on Gender Roles

To measure country-level attitudes on gender roles, I use the World Values Survey (Inglehart et al., 2018), a nationally representative survey which asks attitudinal questions consistently across countries and over time. The most frequently repeated gender-related question across survey waves is: “When jobs are scarce, men have more right to a job than women,” asked from wave 2 (1990-1994) to wave 6 (2010-2014). This question reflects the idea that men’s employment is prioritized over women’s. Lacking a survey question that specifically targets gender roles within marriage, I use this closely correlated question instead.⁵⁴

Figure A6: Move Toward Less Conservative Gender Attitudes



Notes: Share agreeing with “When jobs are scarce, men have more right to a job than women” by country and year.

⁵⁴ Bertrand et al. (2020) find that responses to this question are strongly correlated ($r = 0.88$) with answers to another attitudinal question better capturing distinct societal roles of men and women: “A man’s job is to earn money; a woman’s job is to look after the home and family,” across 23 countries (Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Hong Kong, Iceland, Ireland, Italy, Japan, Korea, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Taiwan, the UK, and the U.S.) in 2002, 2006–2013.

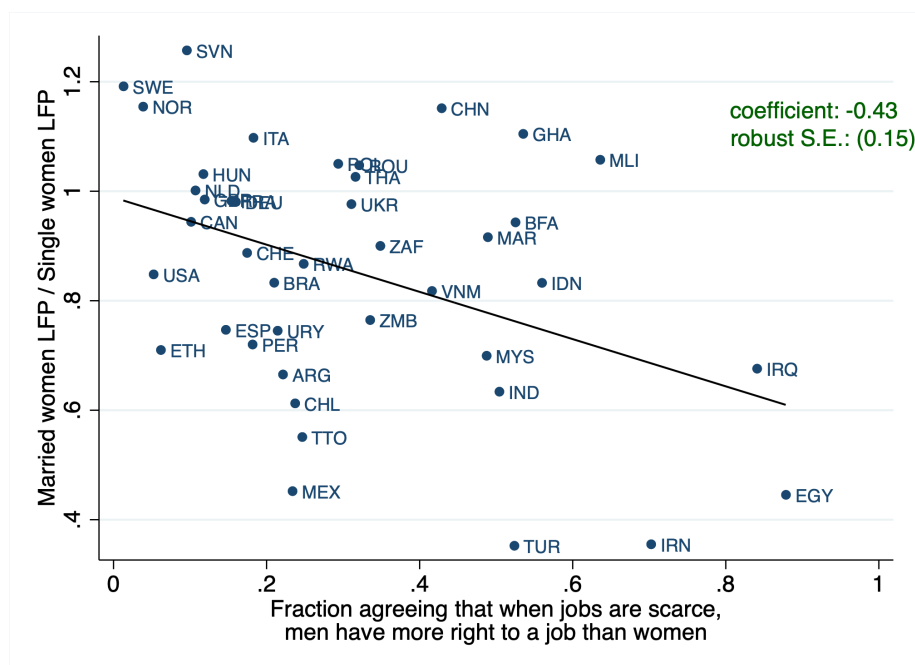
Figure A6 shows the weighted probability of “Agree” by country and year among 20- to 59-year-olds, serving as a proxy for country-level attitudes on gender roles within marriage. Attitudes are becoming less conservative in North America, Europe, and Latin America, while remaining relatively stagnant in Africa and Asia.

C.3 Relationship between married-single FLFP ratio and attitudes

I use the World Values Survey to also compute the labor force participation rate of married and single women, aged 20-59, at the country-wave level. Figure A7 shows that the married-to-single female labor force participation gap is larger in countries with more conservative gender attitudes, mirroring the same correlation observed in the U.S. time series data—married women’s labor force participation catching up with single women’s (Figure 1) as gender attitudes become less traditional (Figure 3).

Although Figure A7 shows a single cross-section (wave 5) for illustration, the negative correlation remains highly significant in a specification that pools waves 2–6 with country and wave fixed effects (coefficient: -0.11 ; standard errors clustered by country: 0.045).

Figure A7: Married Women’s LFP Lags in Countries with Conservative Gender Attitudes

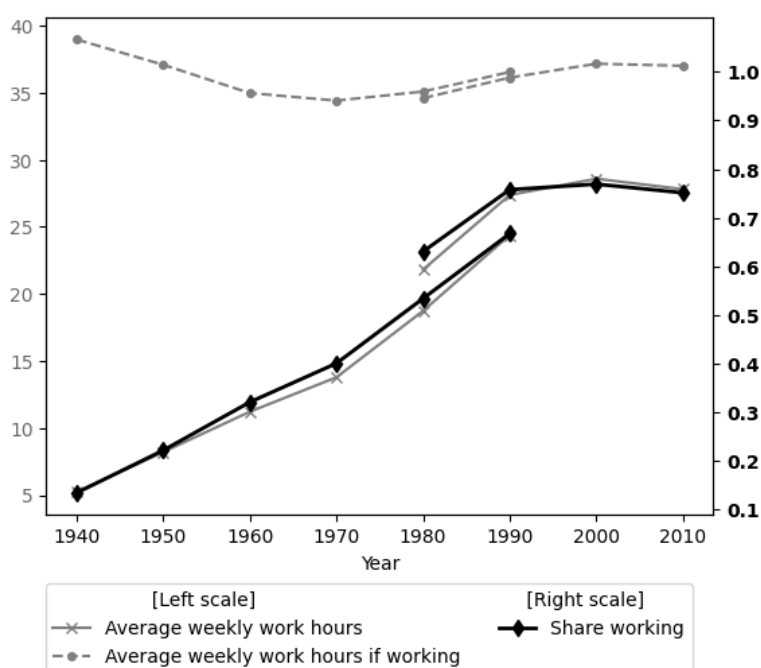


Notes: This figure shows the cross-country correlation between married–single female labor force participation ratios and the conservativeness of gender attitudes. The figure plots the relationship in the World Values Survey Wave 5 (2005-2009), the wave with the largest number of country observations.

Appendix D Growth of Married Women's Market Work

Figure A8 illustrates that the growth in married women's market work between 1940 and 2010 was driven by growth along the extensive margin of labor force participation. It presents the average weekly market work hours of married women, both conditional and unconditional on working, as well as the share of married women working. Conditional work hours (intensive margin) remain stable over time, while the share working (extensive margin) closely tracks the rise in unconditional hours (total growth).

Figure A8: Extensive vs. Intensive Margin of Married Women's Market Work



Note: This figure presents the average weekly market work hours of married women, shown both conditional on working positive hours (circles, dashed lines) and unconditional (crosses, thin solid lines), along with the share of married women working positive hours (diamonds, thick solid lines). Because no single measure of weekly market hours is available across all decades, I plot two overlapping variables to demonstrate that the trend is consistent regardless of the measure used. From 1940 to 1990, I use the midpoint of the intervalled variable *hrswork2* (hours worked last week, intervalled), and from 1980 to 2010, I use *uhrswork* (usual hours worked per week).

In addition to the visual illustration, I formally calculate the contribution of the extensive margin in driving the growth in unconditional work hours.

Average market work hours equals the product of the share of working individuals (extensive margin) and the average hours worked conditional on working (intensive margin):

Where W is the set of individuals who are working,

$$\overline{workhrs} = \frac{\sum_i workhrs_i}{N} = \frac{\sum_{i \in W} workhrs_i}{N} = \frac{|W|}{N} \times \frac{\sum_{i \in W} workhrs_i}{|W|} = \underbrace{\overline{work}}_{\text{extensive}} \times \underbrace{(\overline{hrs|work})}_{\text{intensive}}$$

Thus, the growth in married women's work hours from period s to t can be decomposed into the extensive and intensive margins:

$$\overline{workhrs}_t - \overline{workhrs}_s = \underbrace{(\overline{work}_t - \overline{work}_s) \times \overline{hrs|work}_s}_{\text{extensive margin}} + \underbrace{(\overline{hrs|work}_t - \overline{hrs|work}_s) \times \overline{work}_t}_{\text{intensive margin}}$$

The contribution of the extensive margin to the growth over 1940-2010, then, can be calculated in the following ways.

1. Long-difference: $t = 2010, s = 1940$.

Using a harmonized measure of annual work hours (weekly work hours $\times$ number of weeks worked in Table A8), the result is

$$\frac{(\overline{work}_{2010} - \overline{work}_{1940}) \times \overline{hrs|work}_{1940}}{\overline{workhrs}_{2010} - \overline{workhrs}_{1940}} = 99.1\%$$

2. Total decadal-difference:

Different measures of weekly work hours and the number of weeks worked are used across years to construct the harmonized measure of annual work hours. To circumvent the measurement inconsistency in 1940 and 2010, I also examine the total decadal-difference that uses the same measure within each decadal-difference pair. I use the midpoint of the intervalled variable *hrswork2* (hours worked last week, intervalled) for 1940-1990, and *uhrswork* (usual hours worked per week) for 1980-2010. For 1980-1990, when both variables are available, I take the average of the decadal-difference from each variable. The result is

$$\frac{\sum_{t=1950}^{2010} (\overline{work}_t - \overline{work}_{t-10}) \times \overline{hrs|work}_{t-10}}{\sum_{t=1950}^{2010} \overline{workhrs}_t - \overline{workhrs}_{t-10}} = 96.0\%$$

Therefore, the growth of married women's market work from 1940 to 2010 is almost entirely driven by the extensive margin.

Appendix E Model

E.1 General form of the utility function

The general form of the utility function is given by

$$u_i(Q, C_i, L_f, L_m) = H\left(f(Q)C_i - r(Q)\tau_i w_f L_f + g_i(Q)\right)$$

where the following conditions hold:

Conditions

C1) H is strictly increasing and strictly concave

C2) $(H')^{-1}$ is homogeneous or logarithmically homogeneous^{55,56}

C3) $2p(f')^2 - p \cdot f \cdot f'' + \tau w_f L_f (r'' f' - r' f'') - f' g'' + g' f'' > 0$, where $\tau \equiv \tau_m + \tau_f$, and $g(Q) \equiv g_m(Q) + g_f(Q)$

This general utility function yields the same result that 1) the optimal labor supply decision of the couple is one that maximizes their pooled income less the disutilities from the wife working in the labor market, and 2) the optimal labor decisions are made independently based only on individuals' comparisons of the gains from working in the market versus at home.

⁵⁵Condition C2) is based on [Mazzocco \(2004\)](#), which shows that a collective household's behavior under uncertainty is equivalent to that of a representative agent if and only if H is of the Identically Shaped Harmonic Absolute Risk Aversion (ISHARA) class:

$$-\frac{H_i''(x)}{H_i'(x)} = \frac{1}{\theta x + a_i}$$

⁵⁶CRRA ($H(x) = \ln(x)$, or $H(x) = \frac{x^{1-\theta}-1}{1-\theta}$ for $\theta > 0, \theta \neq 1$) and CARA ($H(x) = -\exp\{-\theta x + b\}$ for $\theta > 0, b \in R$) utility functions—the most commonly used utility functions for risk-averse individuals—satisfy condition C2).

E.2 Model discussion: How home production features as “income” in the budget constraint

It might seem non-standard that according to the budget constraint, private and public goods can be bought with “income” from home production. However, the maximization problem is equivalent to solving

$$\begin{aligned} \max_{Q, C, Y, B} & (Q + Y)(C + D - \tau w_f L_f) \\ \text{s.t.} & \quad pQ + C = w_m L_m + w_f L_f \\ & \quad pY + B = h_m(1 - L_m) + h_f(1 - L_f) \end{aligned}$$

where Y is the non-rival, public component of home production (e.g. cleaning communal area, childcare) and $D \equiv D_m + D_f$ is the total consumption of the private component of home production (e.g. cleaning private space, laundering own clothes). Here, market goods and services Q and C can only be financed from market earnings, while consumption of home-produced goods and services occur within the total home production done by the couple. The market value of private home goods is normalized to be the same as that of the private market good (the numeraire), and the market value of public home goods is the same as the price of public market good.

E.3 Model discussion: value of staying home beyond housework

Following [Gronau \(1977\)](#), I can categorize home production into work at home, which is perfectly substitutable to work in the market, and leisure (i.e. home consumption time), which has poor market substitutes. Then, the utility function—in the general form as in [Appendix E.1](#)—can be formulated as

$$u_i(Q, C_i, L_f, L_m) = H\left(f(Q)C_i - r(Q)[a_i w_i L_i + \tau_i w_f L_f] + g_i(Q)\right)$$

where a_i denotes i ’s preference for leisure/home consumption time, measured proportionally to i ’s market wage.⁵⁷

Ultimately, the optimal labor supply decisions would come down to

$$\begin{aligned} L_m^* &= \mathbb{1}[w_m - a_m \geq h_m] \\ L_f^* &= \mathbb{1}[(1 - \tau)w_f - a_f \geq h_f] \end{aligned}$$

⁵⁷It is easy to also introduce $i \in \{m, f\}$ valuing their spouse’s home consumption time, e.g. the husband valuing the wife’s play time with their children.

E.4 Model extension: Enriching women's labor supply decisions

The independence in the labor supply decisions of husbands and wives simplifies the study of these decisions.⁵⁸ However, a necessary assumption to generate the independence is the perfect substitutability of home-produced goods and market goods. The perfect substitutability assumption, in turn, implies that there are no incentives for specialization. I can enrich the labor supply decisions of married women, however, if I abstract from the labor supply decisions of married men, i.e. have them always work in the market.

As in [Aguiar and Hurst \(2007a\)](#), consider the final consumption good being produced according to a CES production function with market goods and home production as inputs:

$$\left[(y + wL)^\rho + (\underline{h} + h(1 - L))^\rho \right]^{\frac{1}{\rho}} \quad (23)$$

y refers to nonlabor household market income, including any remittances from family, government benefits, and, if married, the husband's earnings. $\underline{h}$ refers to the basic level of home production conducted in the household, including the amount of home production a woman would have completed regardless of her labor force participation, and again, if married, any home production completed by the husband. L is the labor force participation status of single men and women for single households, and of wives for married households.

To make use of the convenient properties of the Fréchet-distributed market and home abilities, a simplification is necessary. A first-order Taylor approximation of equation (23) around $(w = 0, h = 0)$ yields

$$(y^\rho + \underline{h}^\rho)^{\frac{1}{\rho}} + (y^\rho + \underline{h}^\rho)^{\frac{1-\rho}{\rho}} y^{\rho-1} wL + (y^\rho + \underline{h}^\rho)^{\frac{1-\rho}{\rho}} \underline{h}^{\rho-1} h(1 - L) \quad (24)$$

Thus, equation (24) replaces the right-hand side of the budget constraints in household utility maximization problems (equations (2) for married households and (7) for single households). The optimal labor supply decision for single men and women maximizes equation (24):

$$\hat{L}^* = \mathbb{1} [y^{\rho-1} w \geq \underline{h}^{\rho-1} h]$$

A married woman, on the other hand, must consider the cost of the marriage wedge. Where the cost of working in the market is denominated proportionally to the contribution of market goods in

⁵⁸The independence assumption allows me to take advantage of the convenient properties of the Fréchet distribution to model selection into market work versus home production separately by gender. This structure also enables extensions such as Appendix E.5, where I introduce an analogous wedge for married men, motivated by a breadwinner norm that discourages husbands from specializing in home production.

the production of the final good,⁵⁹ her optimal labor supply decision is:

$$L^* = \mathbb{1} \left[(1 - \tau)y^{\rho-1}w \geq \underline{h}^{\rho-1}h \right] \quad (25)$$

The marriage wedge from this extension, comparable to the one in the benchmark model given by equation (20), is:

$$\tau^{qr}(\mathcal{K}) = 1 - \frac{\text{avgwage}_F^{0r}(\mathcal{K})}{\text{avgwage}_F^{qr}(\mathcal{K})} \cdot \underbrace{\left(\frac{\underline{h}_F^{0r}}{\underline{h}_F^{qr}} \cdot \frac{y_F^{qr}}{y_F^{0r}} \right)^{1-\rho}}_{\text{new in the extension}} \cdot \left(\frac{1 - P_F^{0r}(\mathcal{K})}{1 - P_F^{qr}(\mathcal{K})} \right)^{\frac{1}{\theta}} \quad (26)$$

The following describes how I compute the new part of the wedge in this extension.

- ρ : set at 0.5, since the elasticity of substitution between market goods and home produced goods in the literature is estimated at around 2; [McGrattan et al. \(1997\)](#): 1.8, [Chang and Schorfheide \(2003\)](#): 2.3, [Rupert et al. \(1995\)](#): 1.8, [Aguiar and Hurst \(2007a\)](#): 1.78-2.13.
- y : the sum of welfare benefits, social security income, other unearned income, and, if married, spousal earnings in the U.S. Decennial Census. The values are truncated below at 0 and winsorized at the top 10% to remove distortions from outliers, and then I take the average by group. I compute (26) only from 1960 because information on non-wage income becomes available from 1960.
- $\underline{h}$: In the benchmark model, single and married women with the same education and children composition are assumed to have the same home productivity within each decade. Thus, the married-single home productivity ratio is constant over time (fixed at one). In this extension, I impose the analogous restriction that the ratio $\frac{\underline{h}_F^{0r}}{\underline{h}_F^{qr}}$ is constant over time. Because the goal is to compare the evolution trajectory of the marriage wedge in the extension with that in the benchmark model, I set this constant to equate the two wedges in 1960.

⁵⁹The married individual i 's utility in equation (1) becomes $\ln(Q) + \ln(C_i - \tau_i(y^\rho + \underline{h}^\rho)^{\frac{1-\rho}{\rho}} y^{\rho-1} w_f L_f)$.

E.5 Model extension: Adding male marriage wedges

The independence in the labor supply decisions of husbands and wives, arising from the perfect substitutability of home-produced goods and market goods, allows for the addition of *male* marriage wedges. A married couple receives disutility from not only the wife working in the market, but also the husband working in home production. Therefore, married individual $i \in \{m, f\}$ gets the following utility:

$$u_i(Q, C_i, L_f, L_m) = \ln(Q) + \ln(C_i - \tau_i w_f L_f - \tilde{\tau}_i h_m (1 - L_m))$$

where $\tilde{\tau}_i$ represents the disutility that i gets from the husband m working. With $\tilde{\tau} \equiv \tilde{\tau}_m + \tilde{\tau}_f$, the optimal labor supply decisions are:

$$\begin{aligned} L_m^* &= \mathbb{1}[w_m \geq (1 - \tilde{\tau})h_m] \\ L_f^* &= \mathbb{1}[(1 - \tau)w_f \geq h_f] \end{aligned}$$

E.6 Derivation of marriage market equilibrium

The probability that man m chooses spousal type $r \in \{1, \dots, S\}$ or stays single ($r = 0$) is

$$\mathbb{P}(r = \arg \max_{r'=0,1,\dots,S} V_m^{qr'}) = \frac{\exp\{\mathbb{E}(v_m^{qr}) - \pi^{qr} + \psi^{qr}\}}{\sum_{r'=0}^S \exp\{\mathbb{E}(v_m^{qr'}) - \pi^{qr'} + \psi^{qr'}\}}$$

The maximum likelihood estimator of $\mathbb{P}(r = \arg \max_{r'=0,1,\dots,S} V_m^{qr'})$ is the fraction of type q men married to r , or $\frac{n^{qr}}{M^q}$.

Hence, in terms of the number of (q, r) marriages *demanded* by type q men,

$$\ln n^{qr, \mathcal{D}} = \ln n^{q0} + \mathbb{E}(v_m^{qr}) - \pi^{qr} + \psi^{qr} - \mathbb{E}(\hat{v}_m^q)$$

Similarly, woman f of type r choosing her spousal type or remaining single gives the analogue for the number of (q, r) marriages *supplied* by type r women,

$$\ln n^{qr, \mathcal{S}} = \ln n^{0r} + \mathbb{E}(v_f^{qr}) + \pi^{qr} + \psi^{qr} - \mathbb{E}(\hat{v}_f^r)$$

The marriage market-clearing equilibrium transfers π^{qr} are determined such that

$$n^{qr, \mathcal{D}} = n^{qr, \mathcal{S}} = n^{qr}.$$

Appendix F Data

U.S. Census samples The primary data source is the U.S. Decennial Census (Ruggles et al., 2024), accessed through IPUMS USA on October 29, 2024. IPUMS provides multiple samples for each census year. Table A7 indicates the specific sample used for each decade. I use the “default” sample for each year as designated by IPUMS USA. When multiple default samples are available, I select the largest one. However, if the chosen sample exceeds 1% of the population, I randomly draw observations so that each year corresponds to 1% of that year’s population, to ensure consistent representation across years in the pooled data.

Table A7: Raw IPUMS USA Samples Used for each Decade

1900	1%	1930	1%	1960	5%	1990	5% state
1910	1%	1940	1%	1970	1% metro fm2	2000	5%
1920	1%	1950	1%	1980	5% state	2010	ACS 2010, 2011, 2012

Notes: The sample names follow those listed on the IPUMS “Select Samples” page.

Measuring market wages The “market wage” in this paper is defined as each respondent’s total pre-tax wage and salary income for the previous year (variable *incwage*) divided by annual work hours. *incwage* is deflated using the price index variable *CPI99*, so the units are in 2000 US dollars.

Annual market work hours are calculated as the product of weekly work hours and the number of weeks worked, with the underlying variables listed in Table A8. “Usual hours worked per week” is used over “hours worked last week,” if available. Continuous measures are used whenever available. When only interval measures exist, I convert them to continuous values using the corresponding interval averages from other years.

Table A8: Raw Variables for Measuring Annual Market Work Hours by Year

	1940, 1950	1960, 1970	1980, 1990, 2000	2010
Weekly work hours	<i>hrswork1</i>	<i>hrswork2</i>	<i>uhrswork</i>	<i>uhrswork</i>
Number of weeks worked	<i>wkswork1</i>	<i>wkswork2</i>	<i>wkswork1</i>	<i>wkswork2</i>

Notes: *hrswork1*: Hours worked last week; *hrswork2*: Hours worked last week, intervalled; *uhrswork*: Usual hours worked per week; *wkswork1*: Weeks worked last year; *wkswork2*: Weeks worked last year, intervalled.

Appendix G Parameter Inference

G.1 Derivation of the likelihood function to estimate θ

The probability density function of the Fréchet distribution with shape parameter θ and scale parameter s is:

$$f(x; s, \theta) = \frac{\theta}{s} \exp \left\{ - \left(\frac{x}{s} \right)^{-\theta} \right\} \left(\frac{x}{s} \right)^{-\theta-1}$$

The scale parameter in the Fréchet distribution of market abilities among market workers is $\left(\frac{1}{P}\right)^{\frac{1}{\theta}}$, where P is the fraction working in the market among a group defined by gender g , match (q, r) , and children composition $\mathcal{K}$. Therefore, where x_n is the market ability of observation n and P_n denotes the fraction of workers in observation n 's group, the maximum likelihood estimator for θ is:

$$\tilde{\theta}_{MLE} = \arg \max_{\theta \in (0, \infty)} \sum_{n=1}^{Nobs} [\ln \theta - \ln P_n - x_n^{-\theta} P_n^{-1} - (\theta + 1) \ln x_n]$$

G.2 Extracting market abilities from market wages

Recall that an individual i of gender g in a (q, r) match with children composition $\mathcal{K}$ receives wage

$$w_{it} = \bar{w}_{gt}^{qr}(\mathcal{K}) \epsilon_i^w, \quad g \in \{M, F\}$$

where $\bar{w}_{gt}^{qr}(\mathcal{K})$ is the market wage per unit of effective labor for each group. Therefore the x_n in practice equals ϵ_{it}^{w*} where the asterisk denotes that this is the market ability of those who choose to work in the market. Let us isolate ϵ_{it}^{w*} from the observed wages:

$$\text{logwage}_{it} = \ln \bar{w}_{gt}^{qr}(\mathcal{K}) + \ln \epsilon_{it}^{w*}$$

To this end, I regress log wages on (decade $\times$ sex $\times$ education pair $\times$ children composition) dummies. For each group, then, the residuals are

$$\begin{aligned} \text{residuals}_{it} &= \text{logwage}_{it} - \overline{\text{logwage}_{it}} \\ &= (\ln \bar{w}_{gt}^{qr}(\mathcal{K}) + \ln \epsilon_{it}^{w*}) - (\ln \bar{w}_{gt}^{qr}(\mathcal{K}) + \overline{\ln \epsilon_{it}^{w*}}) \\ &= \ln \epsilon_{it}^{w*} - \overline{\ln \epsilon_{it}^{w*}} \end{aligned}$$

Thus,

$$\begin{aligned}
x_n &= \exp\{\text{residuals}_{it} + E(\ln \epsilon_{it}^{w*})\} \\
&= \exp\{\text{residuals}_{it} + \frac{\gamma}{\theta} + \ln(s_n)\} \\
&= \exp\{\text{residuals}_{it} + \frac{\gamma}{\theta} - \frac{1}{\theta} \ln P_n\}
\end{aligned}$$

where γ is the Euler's constant.⁶⁰

G.3 Reformulation of the expected utility from each schooling level

Here, I show how the expected utility from each schooling in equation (16) becomes the version inside the exponential functions in equation (22).

The marriage market clearing condition and equation (4) implies that equilibrium transfers are:

$$\begin{aligned}
\pi^{qr} &= \frac{\ln n^{q0} - \ln n^{0r} - \ln \mu^{qr} - \mathbb{E}(\hat{v}_m^q) + \mathbb{E}(\hat{v}_f^r)}{2} \\
&= \frac{\ln n^{q0} - \ln n^{0r} - \ln \mu^{qr} - 2\mathbb{E}(\ln \hat{Q}_m^q) + 2\mathbb{E}(\ln \hat{Q}_f^r)}{2}
\end{aligned}$$

Then,

$$\begin{aligned}
U_F^r &= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(\mathbb{E}(v_f^{qr}) + \pi^{qr} + \psi^{qr} \right) \right] - c_F^r - \xi^r \\
&= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(2\mathbb{E}(\ln Q^{qr}) + \ln \frac{\mu^{qr}}{1 + \mu^{qr}} + \pi^{qr} + \psi^{qr} \right) \right] + \ln p - c_F^r - \xi^r \\
&= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(2\mathbb{E}(\ln Q^{qr}) + \frac{1}{2} \ln \mu^{qr} - \ln(1 + \mu^{qr}) + \frac{\ln n^{q0} - \ln n^{0r}}{2} - \mathbb{E}(\ln \hat{Q}_m^q) + \mathbb{E}(\ln \hat{Q}_f^r) + \psi^{qr} \right) \right] \\
&\quad + \ln p - c_F^r - \xi^r \\
&= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(2\mathbb{E}(\ln Q^{qr}) + \Psi^{qr} + \frac{\ln n^{q0} - \ln n^{0r}}{2} - \mathbb{E}(\ln \hat{Q}_m^q) + \mathbb{E}(\ln \hat{Q}_f^r) \right) \right] + \ln p - c_F^r - \xi^r \\
&= \sum_{q=0}^S \left[\frac{n^{qr}}{F^r} \left(2A^{qr} + \Psi^{qr} + \frac{\ln n^{q0} - \ln n^{0r}}{2} - \hat{A}_M^q + \hat{A}_F^r \right) \right] + \ln p - c_F^r - \xi^r
\end{aligned}$$

⁶⁰To compute $E(\ln \epsilon_{it}^{w*})$, I need the probability density function of $y = g(x) = \ln(x)$ where x is a Fréchet random variable with scale parameter s and shape parameter θ . $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right| = \frac{\theta}{s} \left(\frac{e^y}{s} \right)^{-\theta-1} e^{-\left(\frac{e^y}{s} \right)^{-\theta}} |e^y|$. Thus, $E(y) = \theta s^\theta \int_{-\infty}^{\infty} y e^{-\theta y} e^{-e^{-\theta y} s^\theta} dy = -\frac{1}{\theta} \int_0^{\infty} e^{-z} (\ln z - \theta \ln s) dz = -\frac{1}{\theta} \Gamma'(1) + \ln s = \frac{\gamma}{\theta} + \ln s$ where γ is the Euler's constant.

It can be shown similarly that

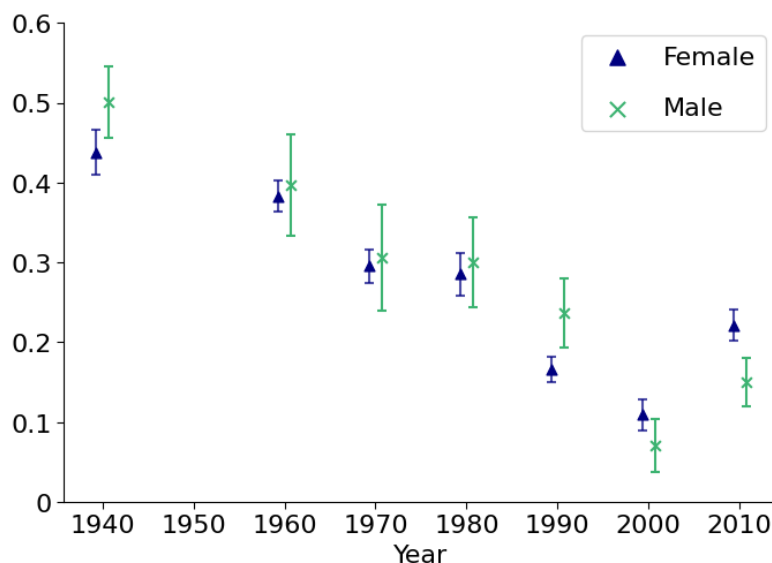
$$\begin{aligned}
U_M^q &= \sum_{r=0}^S \left[\frac{n^{qr}}{M^q} \left(\mathbb{E}(v_m^{qr}) - \pi^{qr} + \psi^{qr} \right) \right] - c_M^q - \xi^q \\
&= \sum_{r=0}^S \left[\frac{n^{qr}}{M^q} \left(2\mathbb{E}(\ln Q^{qr}) + \Psi^{qr} - \frac{\ln n^{q0} - \ln n^{0r}}{2} + \mathbb{E}(\ln \hat{Q}_m^q) - \mathbb{E}(\ln \hat{Q}_f^r) \right) \right] + \ln p - c_M^q - \xi^q \\
&= \sum_{r=0}^S \left[\frac{n^{qr}}{M^q} \left(2A^{qr} + \Psi^{qr} - \frac{\ln n^{q0} + \ln n^{0r}}{2} + \hat{A}_M^q - \hat{A}_F^r \right) \right] + \ln p - c_M^q - \xi^q
\end{aligned}$$

Appendix H Marriage Wedges under Alternative Models

H.1 Male marriage wedges

What if married *men* are also subject to a marriage wedge? The male counterpart to the homemaker gender role would be the breadwinner gender role where married men are expected to earn a market income. This situation corresponds to the model extension in Appendix E.5. Because the optimal labor supply decisions of husbands and wives are independent, the values of the marriage wedge applying to married women are not affected. Nonetheless, it would be interesting to compare the female marriage wedge to the male marriage wedge.

Figure A9: Female and Male Marriage Wedges by Decade



Note: This figure plots the weighted median of τ , the female marriage wedge, and $\tilde{\tau}$, the male marriage wedge (described in Appendix E.5), inferred for each group, by decade. The weight equals the empirical probability of each group. The values of τ here equal the values for the benchmark model in Figure 7. The error bars indicate 95% confidence intervals based on bootstrapped standard errors with 100 replications.

Figure A9 shows that the male marriage wedge declines over time, resembling the trend of the female marriage wedge. The decline indicates that the labor force participation of married men is falling faster than that of comparable single men. In fact, the male wedge is even higher than the female counterpart in 1940 and is lower towards the end of the sample period. This result may be counterintuitive; it is difficult to reconcile the consistent and large decline with the fact that the rise

of the stay-at-home fathers is only a recent phenomenon.

However, the male marriage wedge must be interpreted with a grain of salt. The first reason is an algebraic one. The male wedge is proportional to the ratio of *nonparticipation in the labor force* of married and single men.⁶¹ Throughout the sample period, the male labor force participation rate is very close to 1 regardless of marital status, so that small changes in the labor force participation rate translate to large changes in the marriage wedge. For the same reason, the male marriage wedge is also much noisier than the female counterpart, as depicted by the wider confidence intervals.⁶² The second reason is that nonparticipation is less synonymous with home production for men, making the male marriage wedge harder to interpret substantively. Consistent with this caveat, state-level male marriage wedges are not correlated with state-level attitudes toward gender roles, unlike the female marriage wedges in Table 1.

H.2 Robustness of the marriage wedge

Figure A10 demonstrates the robustness of the marriage wedge to alternative model specifications and sample selection criteria.

“Baseline” refers to the wedge in the baseline model, reproducing the benchmark wedge in Figure 7. The rest are described as follows.

- (1) I incorporate differential tax treatment by marital status by using imputed after-tax wages instead of gross wages. A married woman’s labor supply decision may differ from that of an otherwise similar single woman due to access to more favorable tax arrangements. Because after-tax earnings are not recorded in the U.S. Decennial Census, I apply federal nominal income tax rates from [Tax Foundation \(2013\)](#) to reported before-tax earnings, assuming individuals file under the most beneficial category: Married Filing Jointly, Married Filing Separately, Single, or Head of Household.
- (2) In the baseline, market wages are calculated as wage and salary income divided by annual market work hours (see Appendix F). Here, I augment income by including net pre-tax business, professional practice, and farm income (variable *incbusfm*), available from 1960.
- (3) In the baseline, the Fréchet shape parameter θ , which governs the dispersion of idiosyncratic market and home abilities, is estimated separately for each decade. Here, I estimate a con-

⁶¹The wedge for a married man in a (q, r) match with children composition $\mathcal{K}$ is

$$\tilde{\tau}^{qr}(\mathcal{K}) = 1 - \frac{\text{avgwage}_M^{qr}(\mathcal{K})}{\text{avgwage}_M^{q0}(\mathcal{K})} \left(\frac{1 - P_M^{qr}(\mathcal{K})}{1 - P_M^{q0}(\mathcal{K})} \right)^{\frac{1}{\theta}}$$

⁶²The noisiness reflects the difficulty of estimation when a parameter is near the boundary.

stant θ using pooled data from 1940–2010, following [Hsieh et al. \(2019\)](#), implying that the distribution of innate talent does not change over time.

- (4) The baseline defines married couples as those with a “probable” spouse present in the household, including cohabiting partners, as identified by IPUMS. Here, I adopt a stricter definition, restricting married couples to those in legal marriages—specifically, individuals whose marital status is “married, spouse present in household.”
- (5) To assess the sensitivity of the marriage wedge to the composition of singles, I exclude widows and widowers from the group of single individuals. Widowed singles are a category of interest, as widowhood is more exogenously determined than never marrying or separating.
- (6) A married woman’s labor supply decision may differ from that of a single woman due to higher future childbearing prospects, even when their current children composition index is the same. Childbearing prospects affect a woman’s decision to work, as the expectation of fewer career interruptions increases the incentive to participate in the labor market. Employers may also be less willing to hire women they perceive as likely to leave temporarily or permanently in the future. To account for this difference, I limit the sample to individuals aged 40–54, past the prime childbearing age, rather than 25–54 in the baseline.
- (7) The schooling levels vary by decade in the baseline ([Appendix Table A2](#)), balancing the need for detailed categories to ensure comparability between married and single women with the need for sufficient sample sizes to reduce noise in the estimated marriage wedge. Here, I test robustness to using a fixed set of education levels across all decades. To ensure sufficient sample sizes in each decade, they are: less than high school, high school graduate, and more than high school.

Across these alternative specifications, the magnitude and trajectory of the estimated marriage wedges remain consistent, providing reassurance about the robustness of the baseline results.

Figure A10: Marriage Wedges Under Alternative Specifications or Sample Criteria



Note: This figure plots the weighted median of τ by decade, under alternative model specifications or sample criteria.

Appendix I Counterfactual Steps and Robustness

I.1 Steps for conducting counterfactuals

I denote counterfactual values with underlines.

1. Compute $\underline{A}^{qr}$ and $\underline{P}_g^{qr}(\mathcal{K})$. They differ from the values at the status quo because optimal labor decisions would change under different marriage wedges. Note that $\hat{A}_g^s$ remains unchanged, as singles' labor decision is unaffected by the marriage wedge.
2. Solve for the number of marriage matches $\underline{n}^{qr}$ and education shares $\{\underline{M}^q, \underline{F}^r\}$, by substituting $\underline{A}^{qr}$ with $\underline{A}^{qr}$ in the marriage-market equation (21) and education equation (22). I also use the accounting identities given by equations (12) and (13), and assume that total population size is fixed.
3. Finally, compute $\underline{Y}^{mkt}$ and $\underline{Y}^{home}$.

I.2 Robustness of the main counterfactual

This section describes the different cases in Figure 11, which shows how the results of the main counterfactual (change in aggregate output in 2000 with 1940 marriage wedge) change under alternative assumptions.

I.2.1 Case 2: CES production function with male and female labor

Assume the market production function is no longer linear in male and female labor, but instead takes the Constant Elasticity of Substitution (CES) form, as in Acemoglu et al. (2004). Following their estimation results, the elasticity of substitution, $1 - \frac{1}{r}$, is set to 3: male and female labor are close, but far from perfect, substitutes.

$$Y^{mkt} = B [(1 - \lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{1/r} \quad (27)$$

The male and female wages are given by

$$w_M = B [(1 - \lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{\frac{1-r}{r}} (1 - \lambda)\mathcal{M}^{r-1} \quad (28)$$

$$w_F = B [(1 - \lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{\frac{1-r}{r}} \lambda\mathcal{F}^{r-1} \quad (29)$$

In counterfactual simulations, wages w_M and w_F change depending on how $\mathcal{M}$ and $\mathcal{F}$ change. This is the key difference from the baseline setup with a linear production function.

The wage ratio is

$$\frac{w_F}{w_M} = \frac{\lambda}{1-\lambda} \left(\frac{\mathcal{M}}{\mathcal{F}} \right)^{1-r} \quad (30)$$

For each gender, normalize, as 1 unit of effective labor, the labor provided by singles with schooling level of 1, market ability of 1, and zero children ($\mathcal{K}_0$), i.e. $\bar{w}_M^{10}(\mathcal{K}_0)$ and $\bar{w}_F^{01}(\mathcal{K}_0)$.

$$\begin{aligned} \mathcal{M} &= \sum_{q=1}^S \sum_{r=0}^S \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{w}_M^{qr}(\mathcal{K})}{\bar{w}_M^{10}(\mathcal{K}_0)} \right) \left(P_M^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \\ \mathcal{F} &= \sum_{r=1}^S \sum_{q=0}^S \sum_{\mathcal{K}} n^{qr} d^{qr}(\mathcal{K}) \left(\frac{\bar{w}_F^{qr}(\mathcal{K})}{\bar{w}_F^{01}(\mathcal{K}_0)} \right) \left(P_F^{qr}(\mathcal{K}) \right)^{1-\frac{1}{\theta}} \Gamma \left(1 - \frac{1}{\theta} \right) \end{aligned}$$

To perform counterfactual simulations, I take the following steps. First, the share parameter λ is backed out from equation (30) in the status quo (in status quo, $w_M = \bar{w}_M^{10}(\mathcal{K}_0)$ and $w_F = \bar{w}_F^{01}(\mathcal{K}_0)$):

$$\lambda = \frac{\bar{w}_F^{01}(\mathcal{K}_0) \mathcal{F}^{1-r}}{\bar{w}_F^{01}(\mathcal{K}_0) \mathcal{F}^{1-r} + \bar{w}_M^{10}(\mathcal{K}_0) \mathcal{M}^{1-r}}$$

Next, the exogenous Total Factor Productivity parameter B is backed out from equation (28):

$$B = \frac{\bar{w}_M^{10}(\mathcal{K}_0)}{[(1-\lambda)\mathcal{M}^r + \lambda\mathcal{F}^r]^{\frac{1-r}{r}} \lambda \mathcal{F}^{r-1}}$$

To solve the model, I iterate until convergence.

1. Start with $w_F^{(0)} = \bar{w}_F^{01}(\mathcal{K}_0)$, $w_M^{(0)} = \bar{w}_M^{10}(\mathcal{K}_0)$.
2. Find $\mathcal{M}^{(1)}$, $\mathcal{F}^{(1)}$ when the wages are $w_F^{(0)}$, $w_M^{(0)}$ in the counterfactual (This coincides with the counterfactual in the linear production function case).
3. Find the new wages $w_F^{(1)}$, $w_M^{(1)}$ that are implied by $\mathcal{M}^{(1)}$, $\mathcal{F}^{(1)}$.
4. Repeat steps 2 and 3 until

$$\left| \left(\frac{w_F}{w_M} \right)^{(n)} - \left(\frac{w_F}{w_M} \right)^{(n-1)} \right| \leq 0.0000001$$

Finally, $Y^{mkt} = w_M^{(n)} \times \mathcal{M}^{(n)} + w_F^{(n)} \times \mathcal{F}^{(n)}$.

I.2.2 Case 3: Model extension with income effects

Case 3 follows the description of the model extension in Appendix E.4, where the final consumption good of a household is produced according to the CES production function with market goods and home production as inputs.

The calculation of aggregate market output remains the same as in the baseline, normalized to the average real hourly wage. Aggregate total output, however, is computed as the average of the final consumption good produced in each household (equation (24)), unlike in the baseline, which simply adds the average real hourly wage and the average real hourly home productivity.

In the counterfactual scenario, married women's optimal labor supply changes with τ , as given by equation (25). The household's economic utility also adjusts accordingly, reflecting the revised budget constraint where the final consumption good now appears on the right-hand-side.

The rest of the counterfactual steps proceed in the usual way, described in Appendix I.1.

I.2.3 Cases 4 and 5: Compression of wage differentials

For cases 4 and 5, I compress the observed wage differentials by group in the following way.

- Case 4: The variation in $\bar{w}_g^{qr}(\mathcal{K})$ by marital status and spousal education is suppressed. Denote the new $\bar{w}_g^{qr}(\mathcal{K})$ in case 4 as $\bar{\bar{w}}_g^s(\mathcal{K})$. $\bar{\bar{w}}_g^s(\mathcal{K})$ is set to equal the weighted average of the baseline $\bar{w}_g^{qr}(\mathcal{K})$ for each combination of year, sex, own education level, and family composition, using the empirical share of each group as weights. The same is done for $\bar{h}_M^{qr}(\mathcal{K})$.⁶³
- Case 5: In addition to Case 4, I further scale down the variation in $\bar{w}_g^{qr}(\mathcal{K})$ by own education to 85% of its baseline value. Denote the new $\bar{w}_g^{qr}(\mathcal{K})$ in case 5 as $\tilde{w}_g^s(\mathcal{K})$. I first calculate how much the baseline $\bar{w}_g^{qr}(\mathcal{K})$ increases with own education level on average. To this end, I run a group-level regression of $\bar{w}_g^{qr}(\mathcal{K})$ on a continuous measure of schooling, controlling for year dummies and weighting each group by its empirical share. This regression is run separately for each gender and family composition:

$$\bar{w}_{jt} = \beta^{g\mathcal{K}} \times s_j + \delta_t^{g\mathcal{K}} + \varepsilon_{jt},$$

where $\bar{w}_{jt}$ is the short-hand notation for $\bar{w}_g^{qr}(\mathcal{K})$ of group j with own education level s_j in year t . Then, I start by setting $\tilde{w}_g^1(\mathcal{K}) = \bar{\bar{w}}_g^1(\mathcal{K})$, taken from case 4. Next, I calculate $\tilde{w}_g^s(\mathcal{K}) = \tilde{w}_g^1(\mathcal{K}) + 0.85 \times \hat{\beta}^{g\mathcal{K}} \times (s - 1)$ for $s \geq 2$. The same is done for $\bar{h}_g^{qr}(\mathcal{K})$, the home productivity values of both genders.

⁶³There is no change in women's home productivity $\bar{h}_F^s(\mathcal{K})$, which already does not vary by marital status or spousal education.

Appendix J Wages Gaps by Education: Treatment vs. Selection Effects

The goal of this section is to isolate the role of the treatment effect of acquiring a particular level of education as a driver of the average wage for that level, separate from any selection effects. I use the estimation results in [Chiappori et al. \(2018\)](#).

In [Chiappori et al. \(2018\)](#), $W(\theta_i, s_i, g)$ is the market wage faced by individual i of gender g , education s_i , and ability θ_i . Individuals are categorized into 6 types (human capital levels), from the combination of 3 education levels—secondary ($s = 1$), high school ($s = 2$), and university ($s = 3$)—and 2 ability levels—high ($\theta = H$) and low ($\theta = L$). $W(\theta_i, s_i, g)$ is estimated as

$$\ln W(\theta_i, s_i, g) = \alpha^g \cdot \mathbb{1}(\theta_i = L, s_i = 1) + \beta_2^g \cdot \mathbb{1}(s_i = 2) + \beta_3^g \cdot \mathbb{1}(s_i = 3) + \gamma_{s_i}^g \cdot \mathbb{1}(\theta_i = H) \quad (31)$$

where α^g indicates the gender-specific base wage for a low-ability individual with secondary education, i.e. $\ln W(\theta_i = L, s_i = 1, g)$; β_s^g indicates the gender-specific education premium of acquiring education level $s \in \{2, 3\}$; and γ_s^g indicates the gender $\times$ education-specific ability premium for $s \in \{1, 2, 3\}$. The following estimation results are used as ingredients in my calculation.

- Table 7: $\mathbb{P}((\text{husband's type}, \text{wife's type})|\text{married})$
- Table 6: $\mathbb{P}(\text{type}|\text{single}, g)$
- Table 11: $\mathbb{P}(\text{married}|g)$
- Table 9: $\mathbb{P}(s|g)$
- Table 13: Education premiums, ability premiums, base wages, i.e. $\hat{\alpha}^g, \hat{\beta}_2^g, \hat{\beta}_3^g, \hat{\gamma}_1^g, \hat{\gamma}_2^g, \hat{\gamma}_3^g$

I follow the following steps. First, calculate $\mathbb{P}(\text{type}|\text{married}, g)$ by summing $\mathbb{P}((\text{husband's type}, \text{wife's type})|\text{married})$ over all possible types of the spouse. Next, the share of types by gender is

$$\mathbb{P}(\text{type}|g) = \mathbb{P}(\text{type}|\text{married}, g) \times \mathbb{P}(\text{married}|g) + \mathbb{P}(\text{type}|\text{single}, g) \times \mathbb{P}(\text{single}|g)$$

I then use $\mathbb{P}(\text{type}|g)$ to calculate, for each gender, the share of high-ability individuals in each education level:

$$\mathbb{P}(H|s, g) = \frac{\mathbb{P}(H, s|g)}{\mathbb{P}(H, s|g) + \mathbb{P}(L, s|g)}$$

The average predicted wage by education for each gender takes into account the education premium and the ability premium, following equation (31), as well as the share of high-ability indi-

viduals:

$$\mathbb{E}(\widehat{W}|s, g) = e^{\ln \widehat{W}(H, s, g)} \times \mathbb{P}(H|s, g) + e^{\ln \widehat{W}(L, s, g)} \times \mathbb{P}(L|s, g)$$

Denote as $\tilde{W}$ the predicted wage that includes just the education premium, while excluding the role of ability. Then,

$$\mathbb{E}(\tilde{W}|s, g) = \begin{cases} e^{\hat{\alpha}^g}, & \text{if } s = 1 \\ e^{\hat{\alpha}^g + \hat{\beta}_s^g}, & \text{if } s = 2, 3 \end{cases}$$

Table A9 reports the values of $\mathbb{E}(\widehat{W}|s, g)$ and $\mathbb{E}(\tilde{W}|s, g)$.

Table A9: Wage by Education

Education level	Male		Female	
	Total	Only edu premium	Total	Only edu premium
1	22.68	20.41	15.51	11.60
2	24.17	21.85	15.74	14.84
3	23.89	22.44	20.90	19.11

Note: This table reports two types of wages (earnings, actually, in thousands of 2008 GBP) by gender. “Total” indicates the total wage by education, i.e. $E(\widehat{W}|s, g)$. “Only edu premium” indicates the part of wage that excludes the role of ability, i.e. $E(\tilde{W}|s, g)$.

Finally, Table A10 reports the role of the education premium in accounting for the total wage gap between any two education levels s and s' :

$$\frac{E(\tilde{W}|s', g) - E(\tilde{W}|s, g)}{E(\widehat{W}|s', g) - E(\widehat{W}|s, g)}$$

The education premium accounts for a large share of the total wage gap by education. The smallest (positive) share is 83%. Shares often exceed 100%, which occurs when the ability premium is higher at a lower education level than at a higher one, or when the proportion of high-ability individuals is greater among those with lower education than among those with higher education. If the extent of these effects is large, the average wage can actually decline with higher education, which leads to a negative value for the education premium’s share.

Table A10: Education Premium's Role in Wage Gap by Education

	Compared education levels		
	2 vs. 1	3 vs. 2	3 vs. 1
Male	96%	-209%	168%
Female	1372%	83%	139%

Note: This table reports the share of the wage gap by education that is accounted for by the education premium, for any two education levels.

Comparison with Returns-to-Schooling Literature Interestingly, the results of this exercise mirror empirical estimates of ability bias in the returns to schooling. In the pioneering work of [Griliches \(1977\)](#), adding a control for IQ reduces the OLS estimate of the returns to schooling from 6.8% to 5.9%, suggesting that selection effects account for about 13% of the wage increase associated with an additional year of schooling. This is in a similar range to the compression of educational wage differentials by 15% that I adopt in case 5 of Figure 11 to remove the selection effect of education.

However, [Griliches \(1977\)](#) and many others, as surveyed in [Card \(1995\)](#), find that IV estimates of the returns to schooling typically exceed the OLS estimates. This pattern is consistent with some shares exceeding 100% in Table A10. It also suggests that the education premium might even be larger than the observed wage differentials by education, in which case the fall in output in the main counterfactual would be even larger than in the baseline.